SFU NatLangLab

CMPT 713: Natural Language Processing

Pretraining Language Models

Spring 2023

2023-03-23

Pretraining and task-specific fine-tuning

Pretraining

- Big pile of unlabeled text data!
- Lots of resources to train!

Helps to build

- Useful representations of language
- Provide good initial parameters for downstream tasks
- Probability distributions that can be sampled from

Task-specific fine-tuning

- Annotated data specific to a task (usually small)
- Initialize with pre-trained model

Pretraining language models

- Model (Neural Architecture)
  - Does it use FFN, RNN (LSTM, GRU), or Transformer?
    - Is it an **encoder**-based, **decoder**-based, or **encoder-decoder** model?
  - Specifics of the neural architecture (number of layers, embedding size, etc)

- Dataset
  - What is the data that is used to pretrain the model?

- Training objective
  - What is the training objective?

- Other details
  - Tokenization: what tokenization is applied?
  - Implementation and training details?

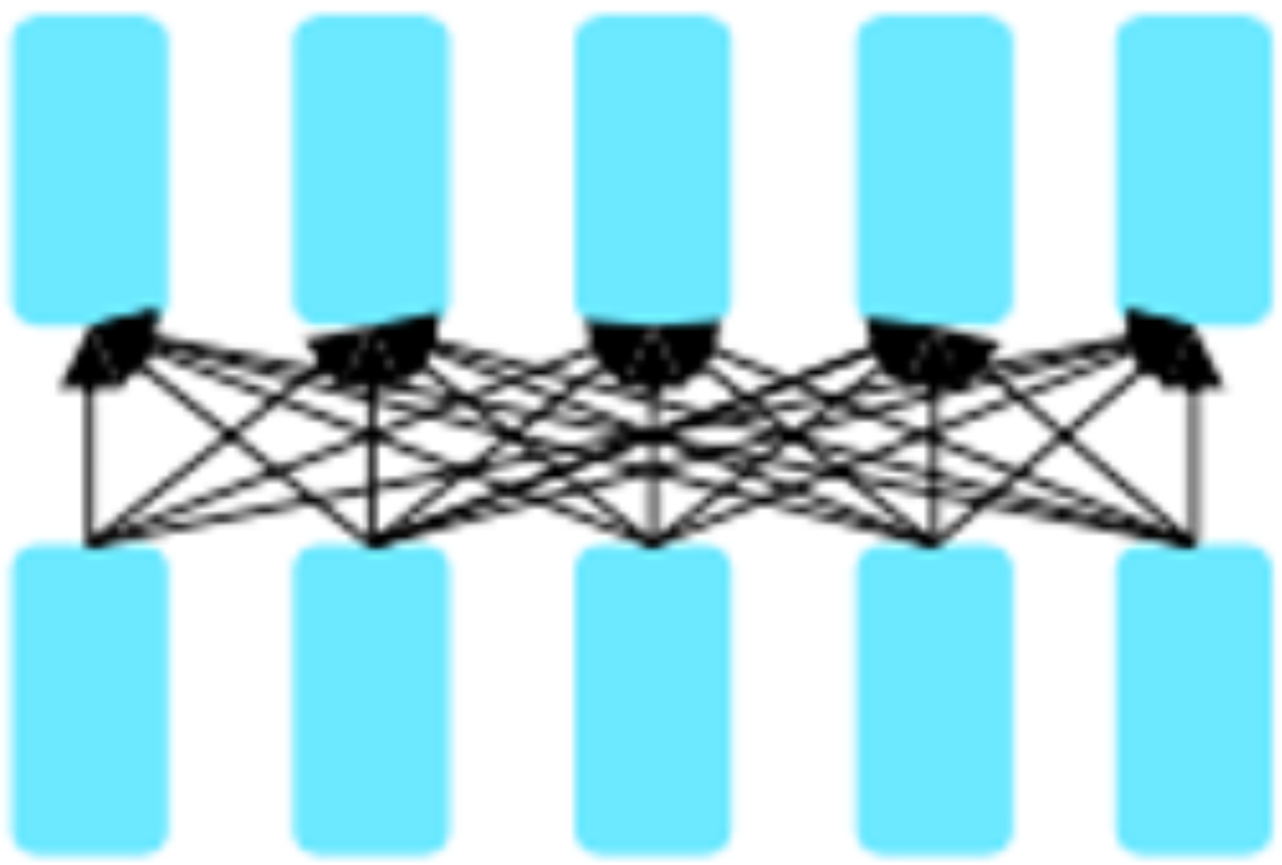
Summary of pretrained models we looked at

Paper	Model	Dataset	Training Objective
W2V CBOW [Miklov et al, 2013]	FFN	Google News (100B words)	Masked LM (within window)
ELMo [Peters et al, 2018]	Bi-LSTM	1B Word benchmark (800M words)	Bidirectional LM
BERT [Devlin et al, 2018]	Transformer (encoder block)	BookCorpus + English Wikipedia (3.3B words)	Masked LM Next sentence prediction

Transformers for pretraining

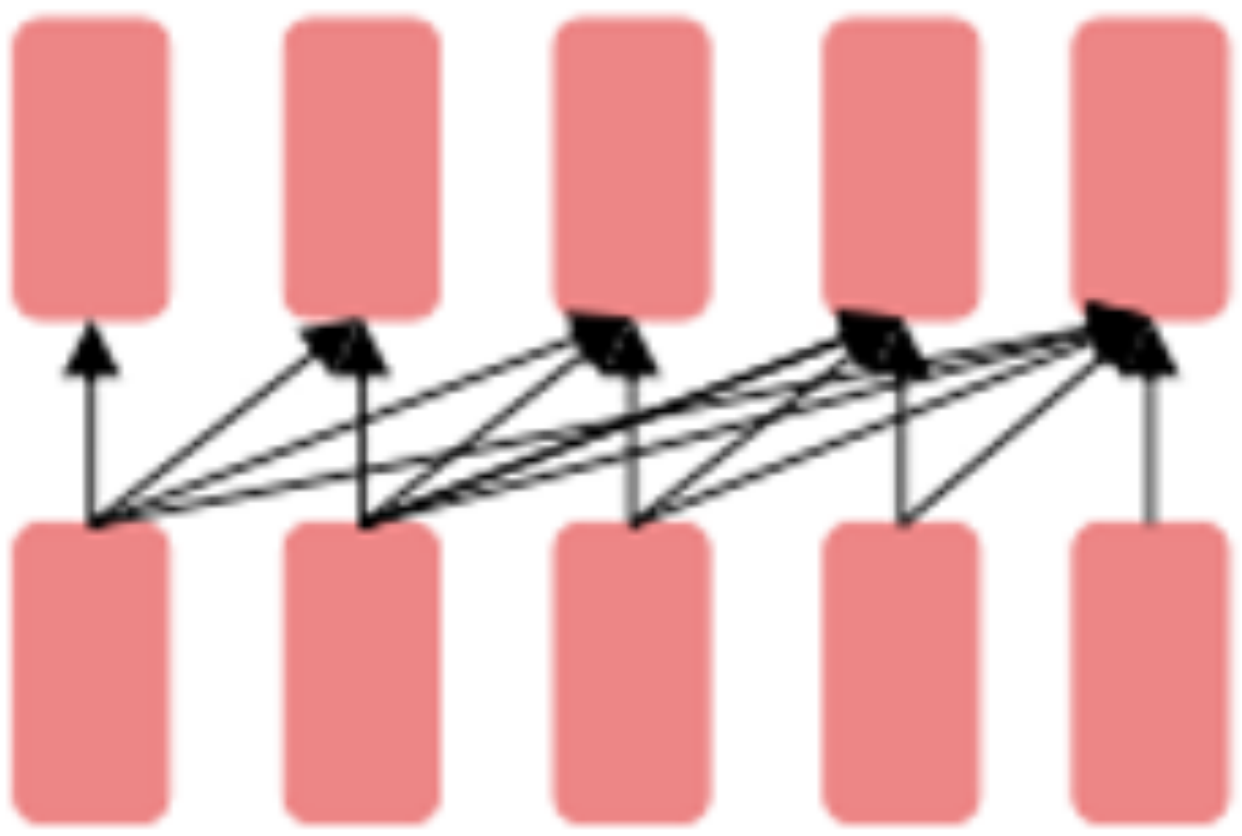
- Self-supervised Transformer based models shattered language understanding benchmarks in NLP in 2018.
- Trained on large text corpus with self-supervised objectives and then transferred.

Encoder only



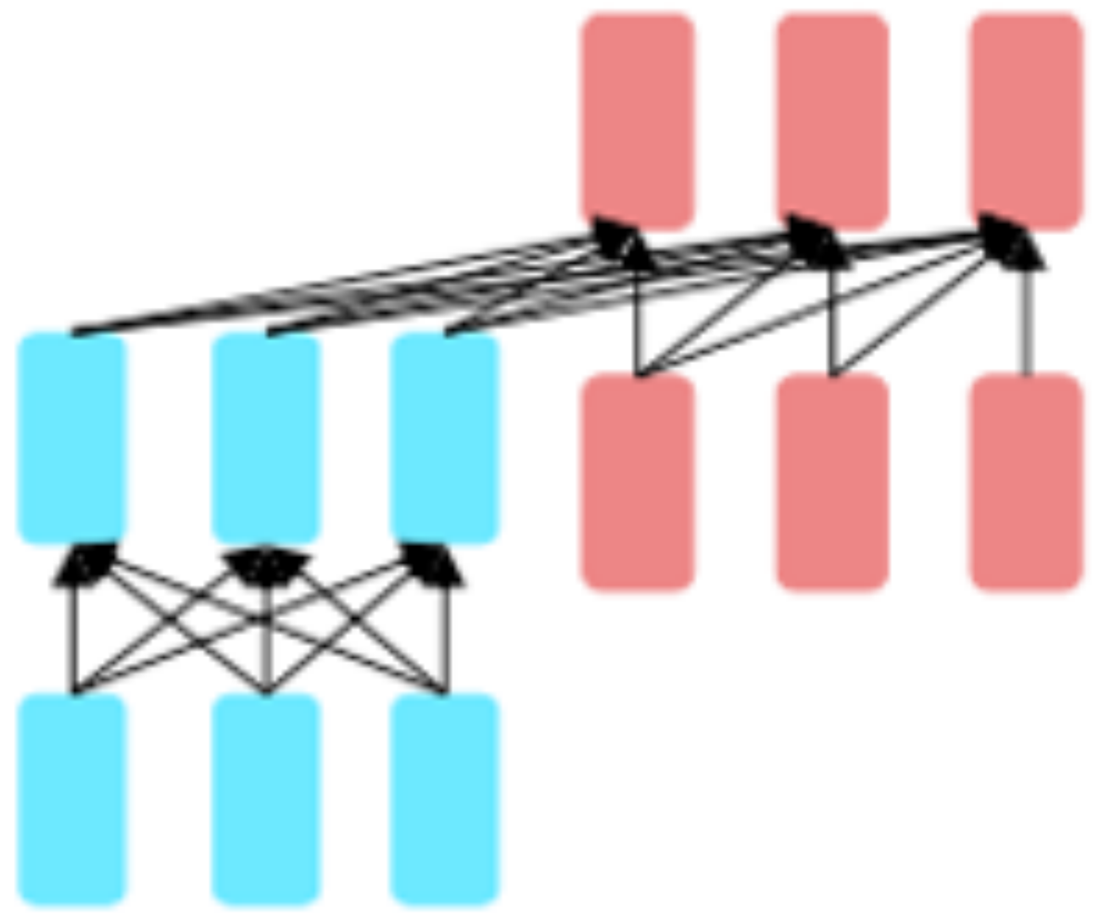
- Masked language models
- Bidirectional context
- BERT + variants (e.g. RoBERTa)

Decoder only



- Language models
- Can't condition on future words, good for generation
- GPT-2, GPT-3, LaMDA

Encoder-Decoder

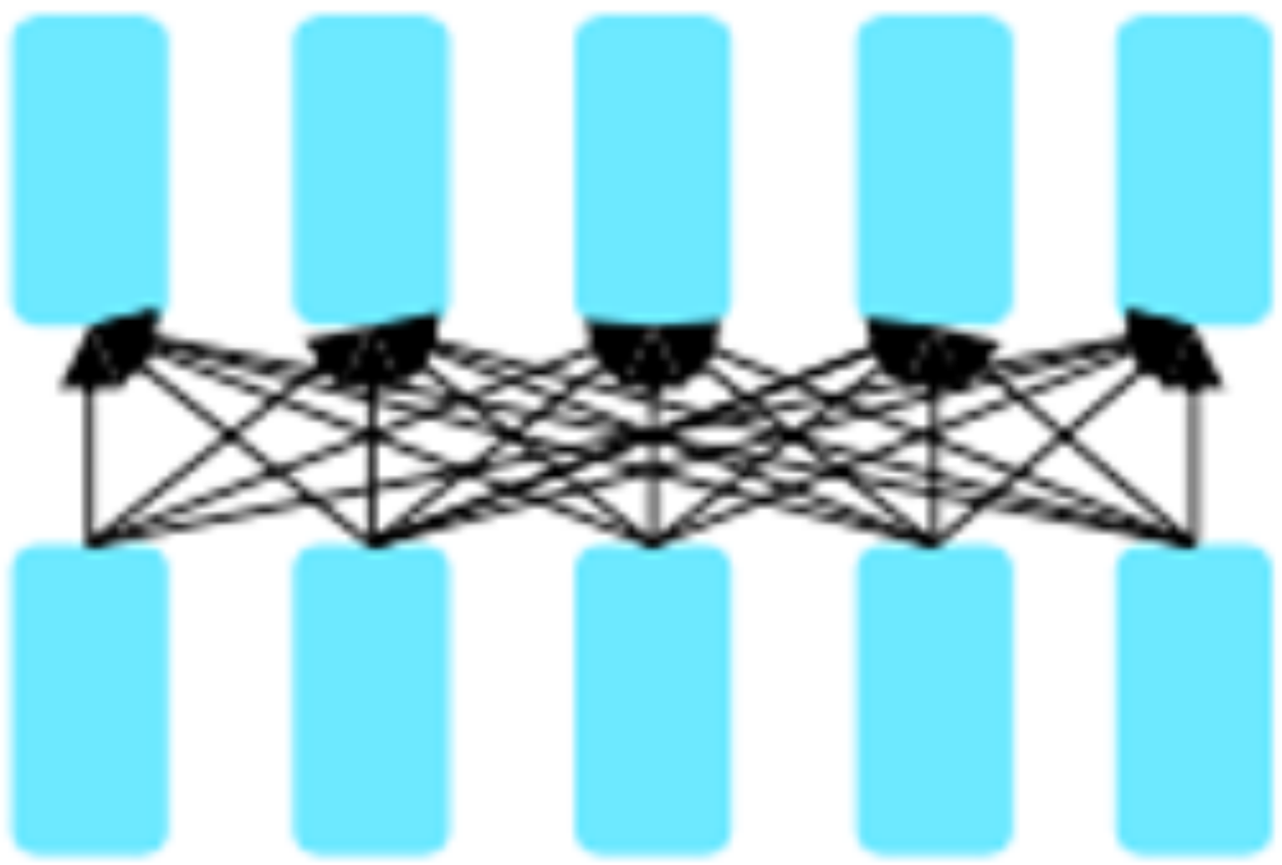


- Combine benefits of both
- Original Transformer, UniLM, BART, T5, Meena

Transformers for pertaining

- Self-supervised Transformer based models shattered language understanding benchmarks in NLP in 2018.
- Trained on large text corpus with self-supervised objectives and then transferred.

Encoder only



- Masked language models
- Bidirectional context
- BERT + variants (e.g. RoBERTa)

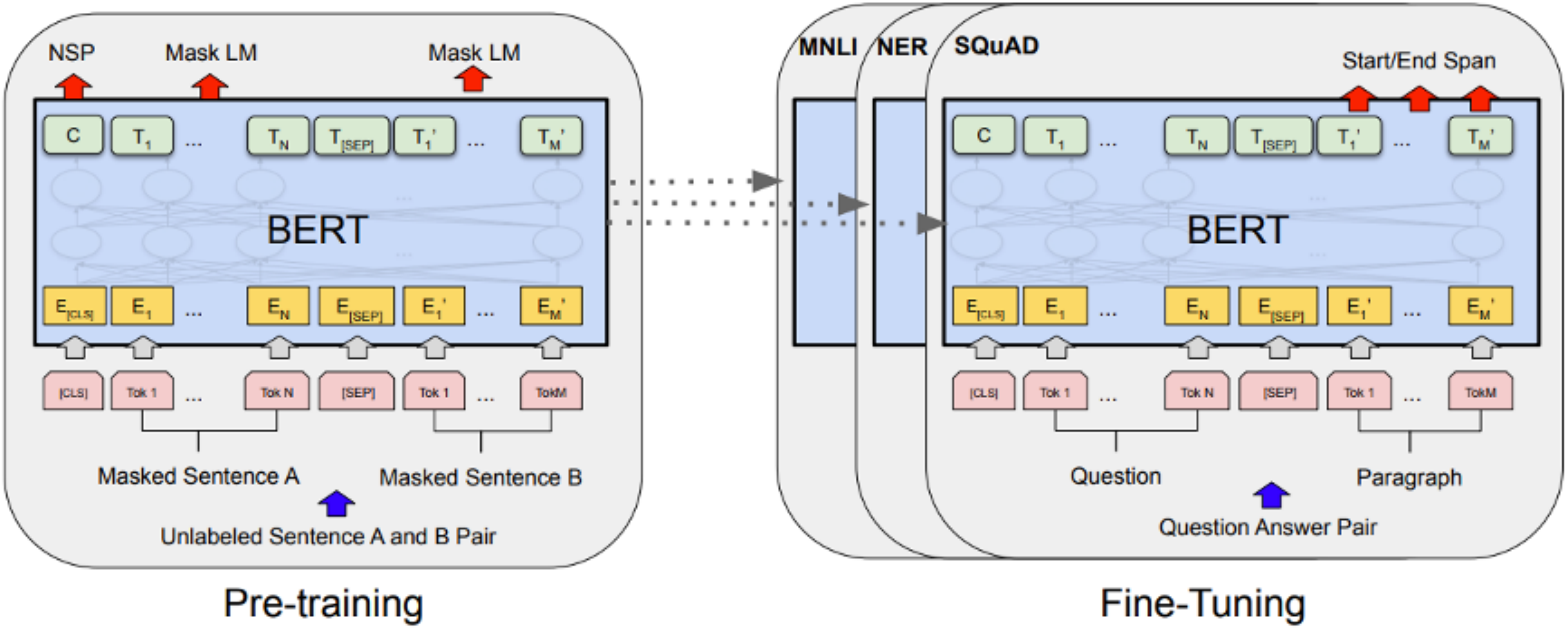
Decoder only

- Language models
- Can't condition on future words, good for generation
- GPT-2, GPT-3, LaMDA

Encoder-Decoder

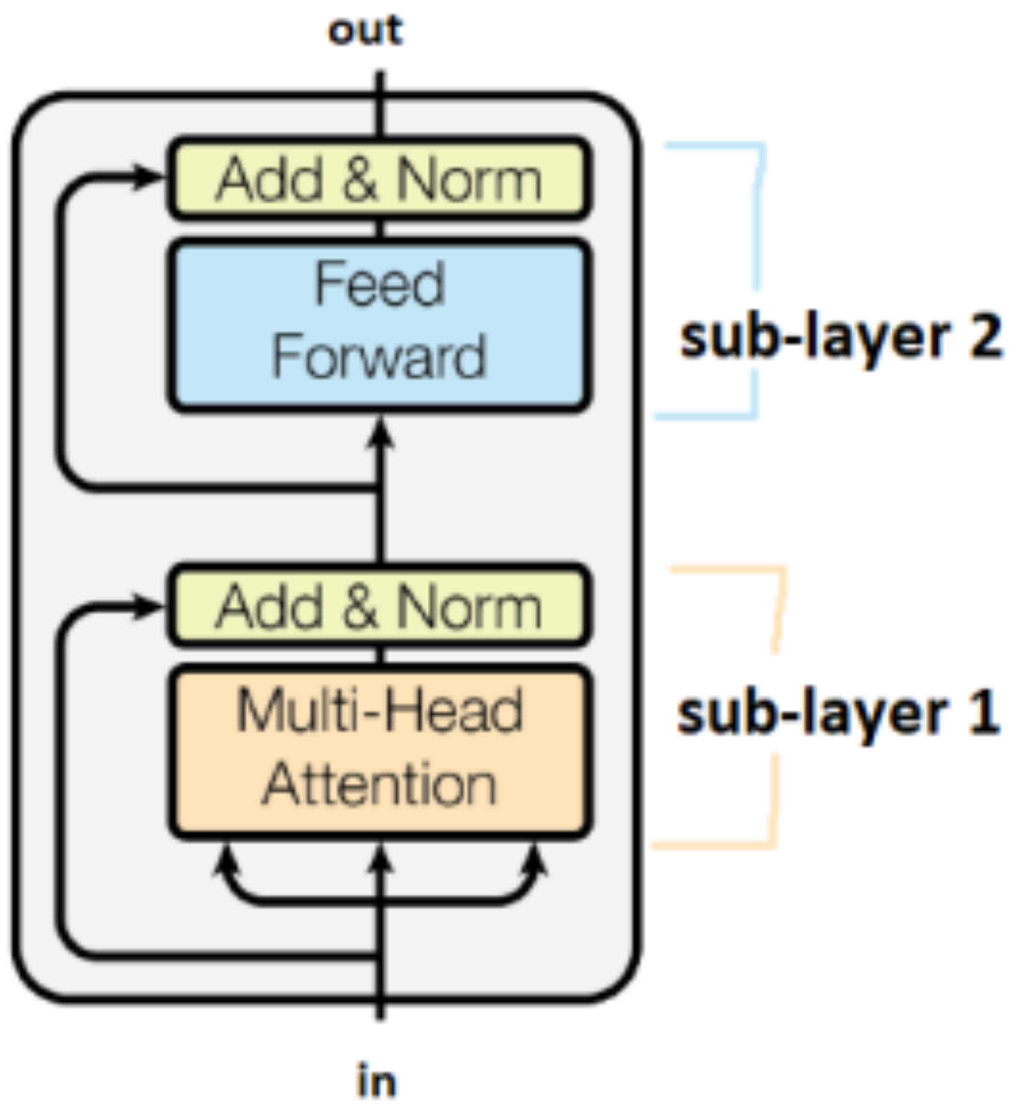
- Combine benefits of both
- Original Transformer, UniLM, BART, T5, Meena

Pre-training and fine-tuning

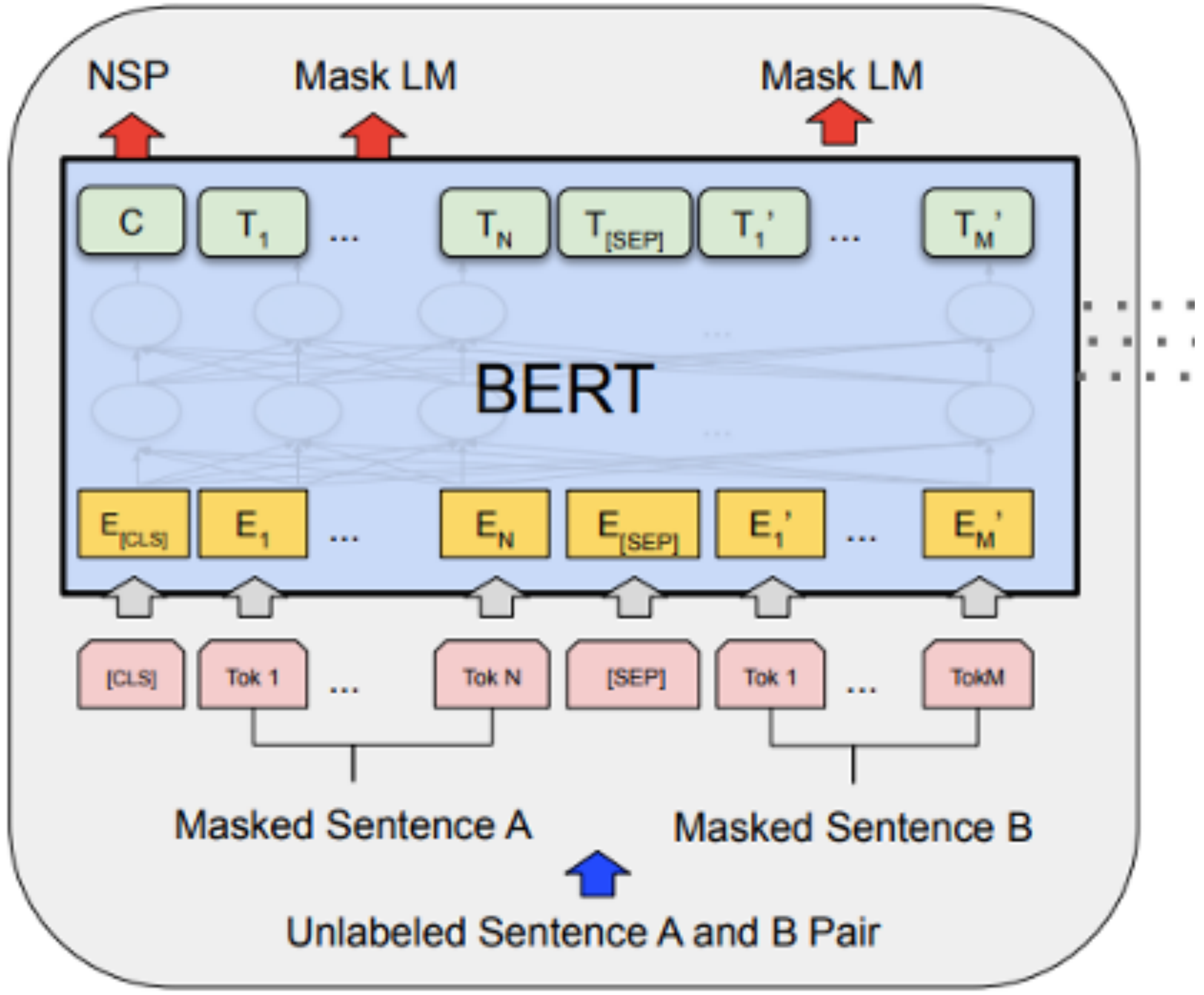


BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding https://arxiv.org/pdf/1810.04805.pdf

BERT



- Transformer Encoder
- Two training objectives
  - Masked Language Modeling
  - Next Sentence Prediction



Pre-training

Masked language models (MLMs)

Mask 15% of tokens

Example: `my dog is hairy`, we replace the word hairy

- 80% of time: replace word with `[MASK]` token

  `my dog is [MASK]`

- 10% of time: replace word with random word

  `my dog is apple`

- 10% of time: keep word unchanged to bias representation toward actual observed word

  `my dog is hairy`

RoBERTa

- Train with more data and for more epochs
  - Vocabulary size of 50K subword units vs 30K for BERT
  - Larger batch size and more training data
- No need for NSP

Model	data	bsz	steps	SQuAD (v1.1/2.0)	MNLI-m	SST-2
RoBERTa						
with BOOKS + WIKI	16GB	8K	100K	93.6/87.3	89.0	95.3
+ additional data (§3.2)	160GB	8K	100K	94.0/87.7	89.3	95.6
+ pretrain longer	160GB	8K	300K	94.4/88.7	90.0	96.1
+ pretrain even longer	160GB	8K	500K	**94.6/89.4**	**90.2**	**96.4**
$\text{BERT}_{\text{LARGE}}$						
with BOOKS + WIKI	13GB	256	1M	90.9/81.8	86.6	93.7

pretrain with **1024 V100 GPUs** for ~1 day

RoBERTa: A Robustly Optimized BERT Pretraining Approach
Liu et al, UW and Facebook, arXiv 2019

RoBERTa

- Train with more data and for more epochs
  - Vocabulary size of 50K subword units vs 30K for BERT
  - Larger batch size and more training data
- No need for NSP

Dynamic masking (masking changes)

Masking	SQuAD 2.0	MNLI-m	SST-2
reference	76.3	84.3	92.8
Our reimplementation:			
static	78.3	84.3	92.5
dynamic	78.7	84.0	92.9

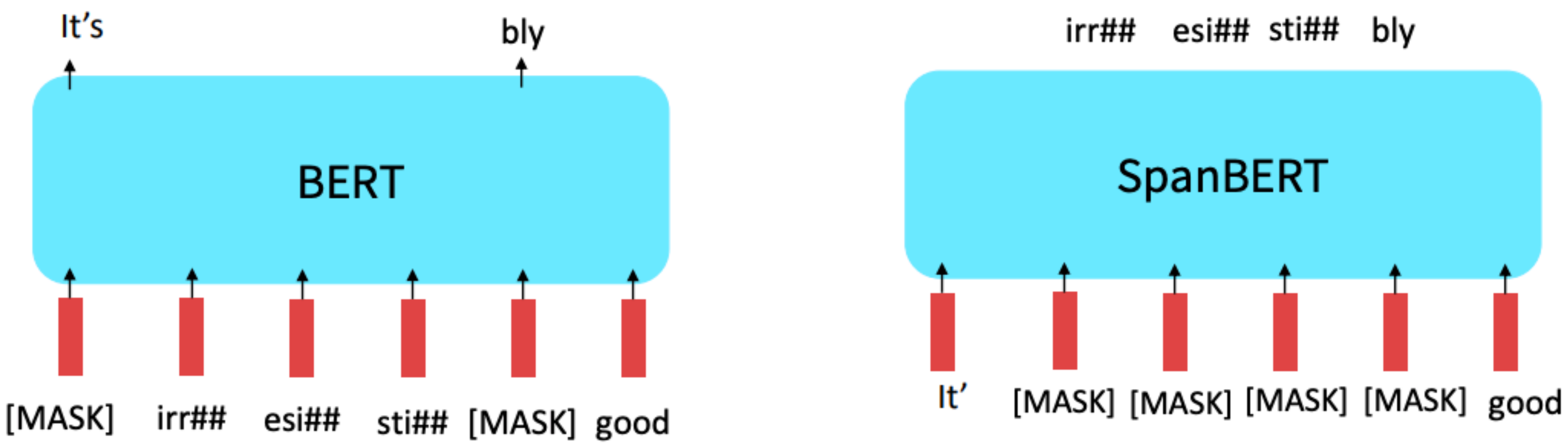
Better results with careful reimplementation.
Mean over 5 random seeds.

Model	SQuAD 1.1/2.0	MNLI-m	SST-2	RACE
Our reimplementation (with NSP loss):				
SEGMENT-PAIR	90.4/78.7	84.0	92.9	64.2
SENTENCE-PAIR	88.7/76.2	82.9	92.1	63.0
Our reimplementation (without NSP loss):				
FULL-SENTENCES	90.4/79.1	84.7	92.5	64.8
DOC-SENTENCES	90.6/79.7	84.7	92.7	65.6
$\text{BERT}_{\text{BASE}}$	88.5/76.3	84.3	92.8	64.3
$\text{XLNet}_{\text{BASE}}$ (K = 7)	–/81.3	85.8	92.7	66.1
$\text{XLNet}_{\text{BASE}}$ (K = 6)	–/81.0	85.6	93.4	66.7

RoBERTa: A Robustly Optimized BERT Pretraining Approach
Liu et al, UW and Facebook, arXiv 2019

SpanBERT

- Mask out spans!



	NewsQA	TriviaQA	SearchQA	HotpotQA	Natural Questions	Avg.
Google BERT	68.8	77.5	81.7	78.3	79.9	77.3
Our BERT	71.0	79.0	81.8	80.5	80.5	78.6
Our BERT-1seq	71.9	80.4	84.0	80.3	81.8	79.7
SpanBERT	**73.6**	**83.6**	**84.8**	**83.0**	**82.5**	**81.5**

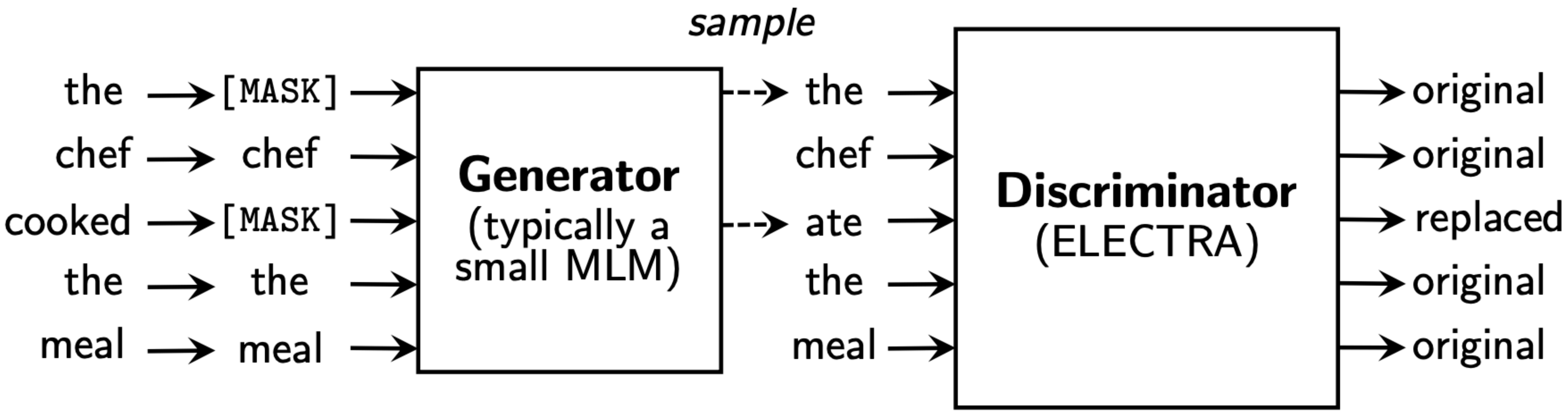
Table 2: Performance (F1) on the five MRQA extractive question answering tasks.

SpanBERT: Improving Pre-training by Representing and Predicting Spans
Joshi et al, TACL 2019

Discriminative training

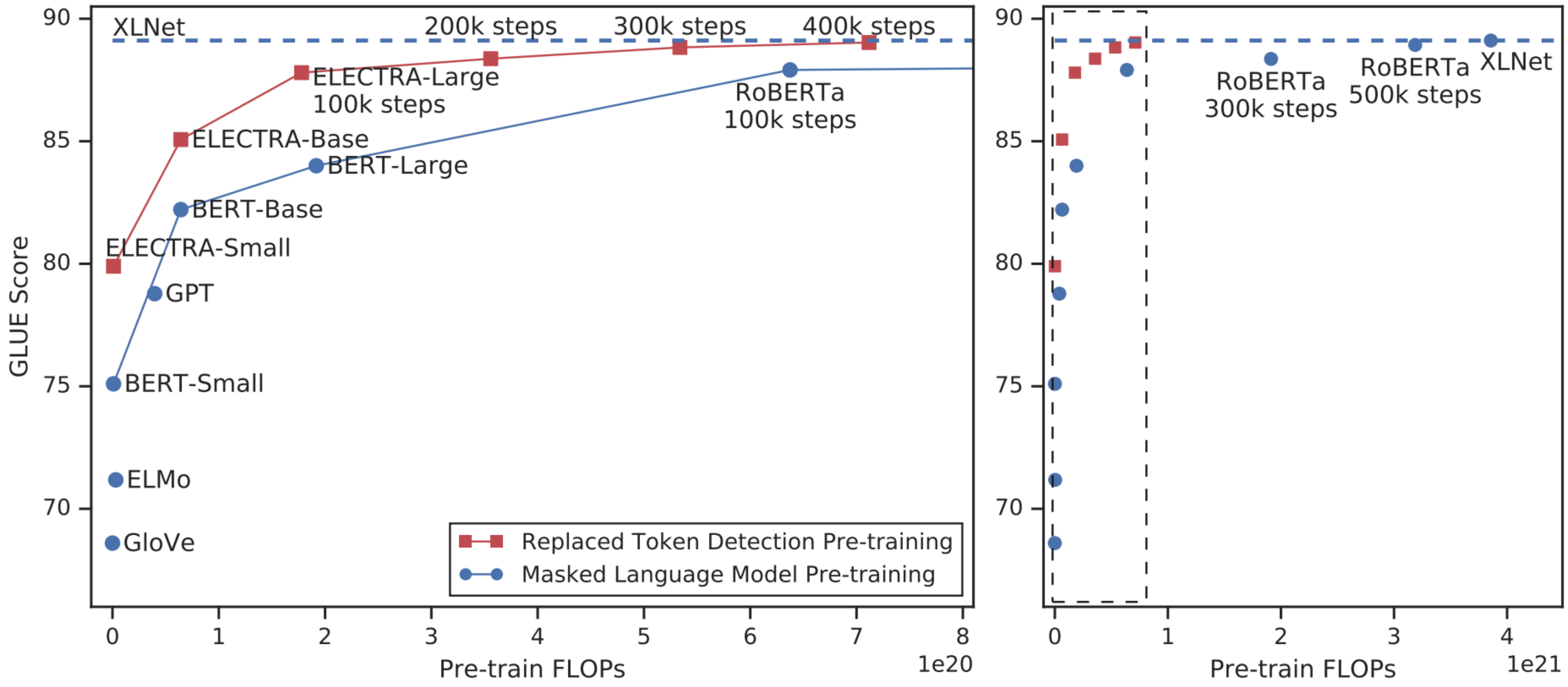
Loss is on all the training tokens vs just the masked ones, more compute efficient use of the training data

Train model to discriminate locally plausible text from real text



ELECTRA: Pre-training Text Encoders as Discriminators Rather Than Generators
Clark et al, ICLR 2020

Discriminative training



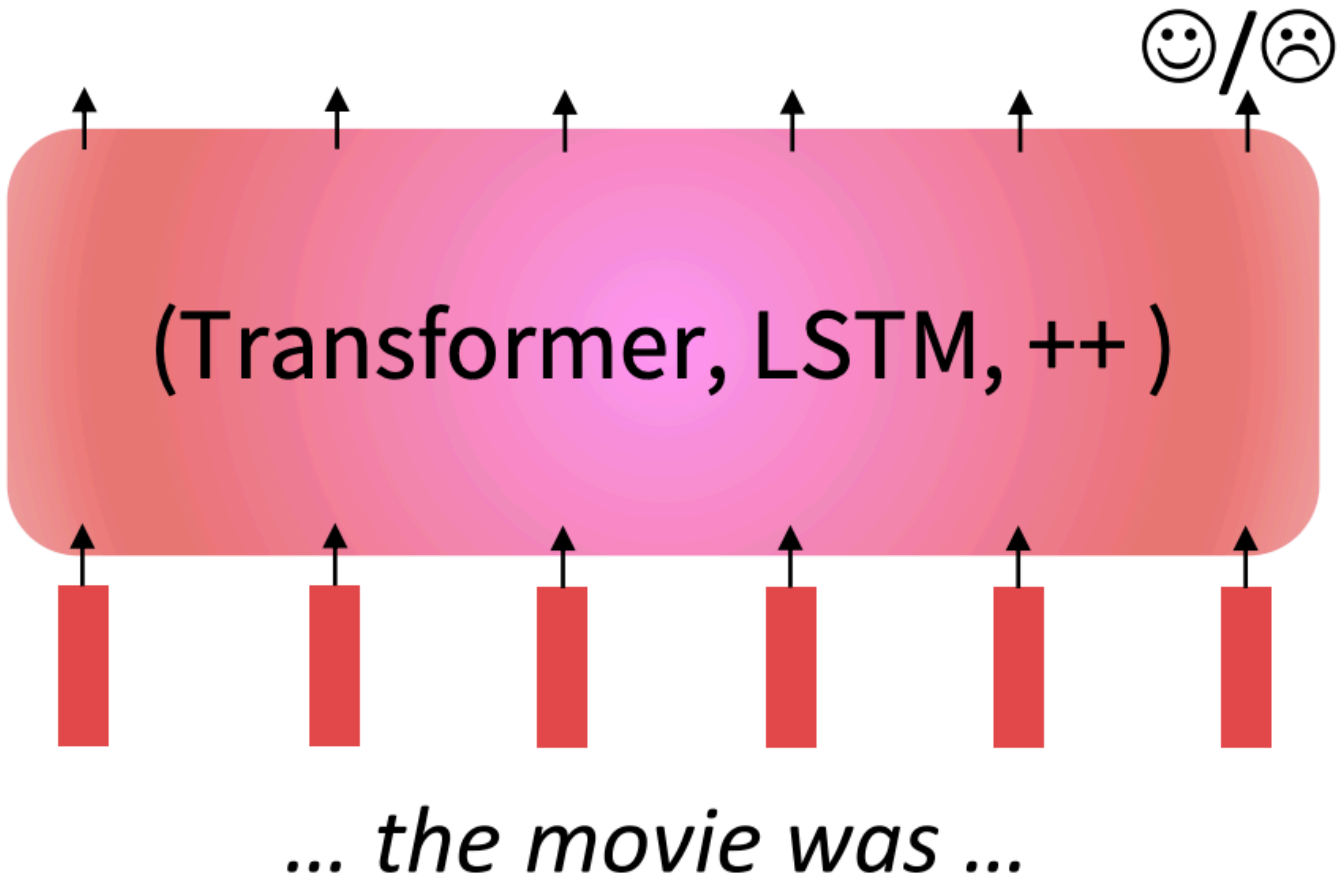
ELECTRA: Pre-training Text Encoders as Discriminators Rather Than Generators
Clark et al, ICLR 2020

Full finetuning vs parameter efficient fine-tuning

- Finetuning every parameter in a pretrained model works well, but is memory-intensive.
- **Lightweight** finetuning methods adapt pretrained models in a constrained way.
- Leads to **less overfitting** and/or **more efficient finetuning and inference**.

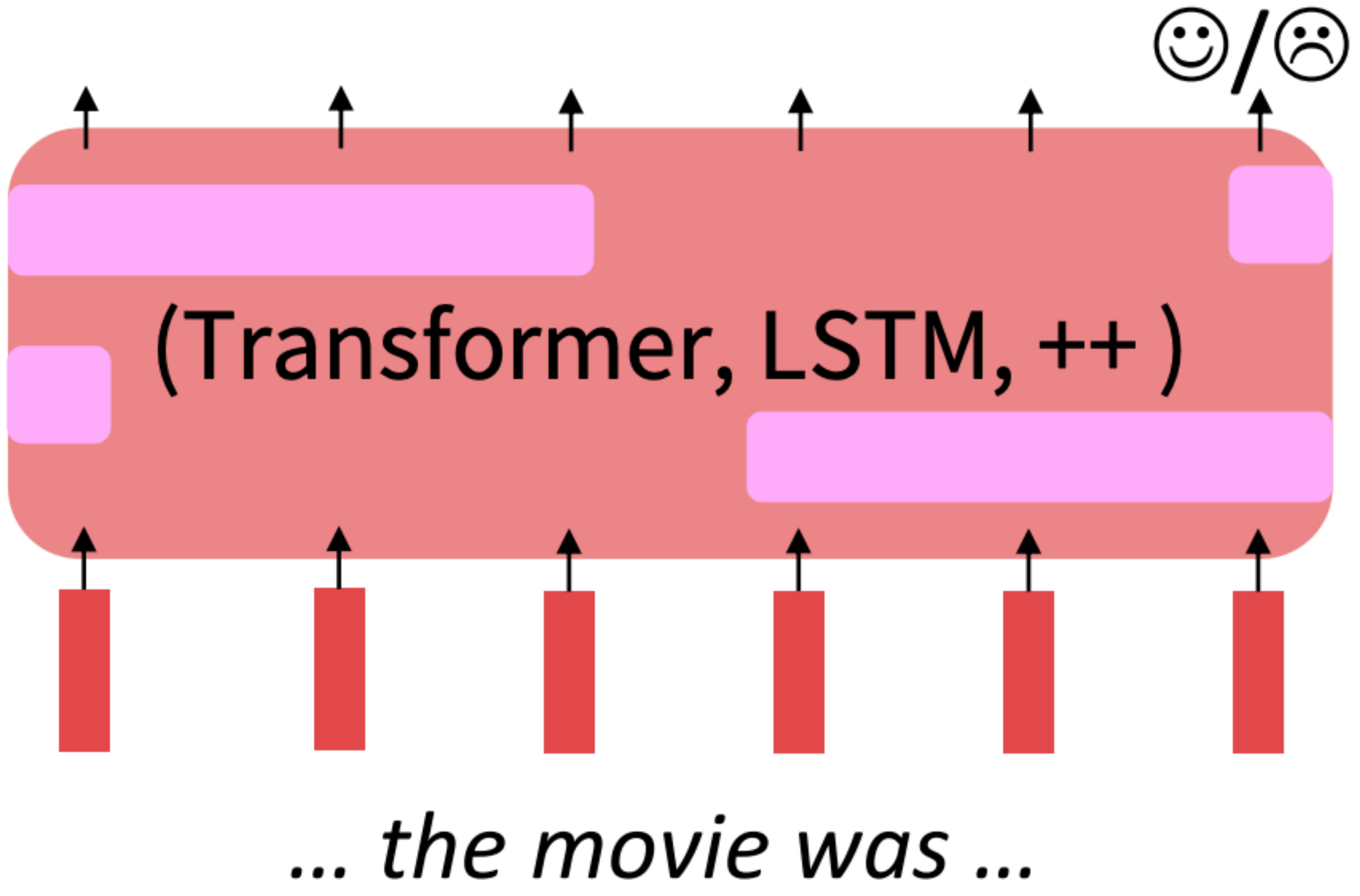
Full Finetuning

Adapt all parameters



Lightweight Finetuning

Train a few existing or new parameters

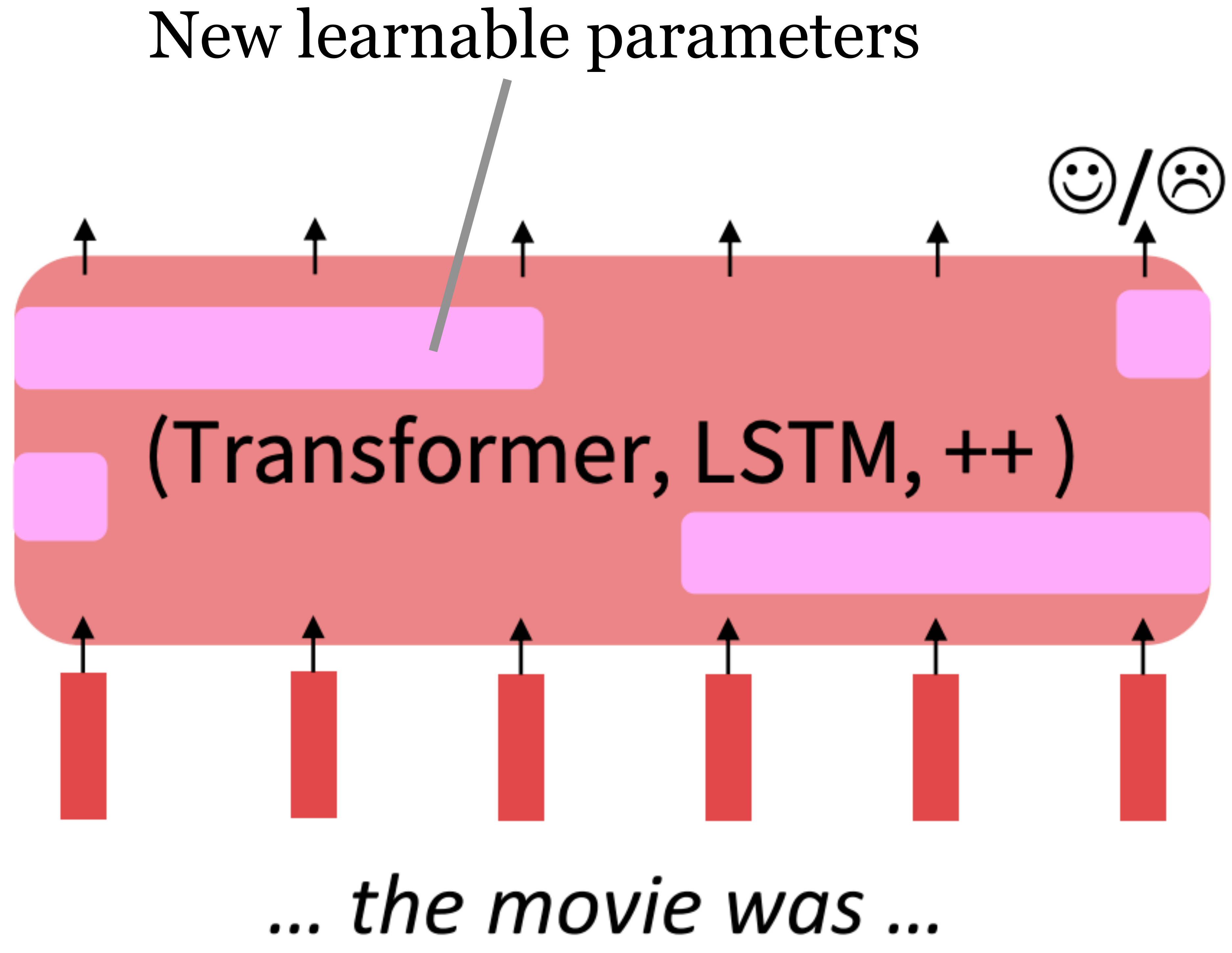


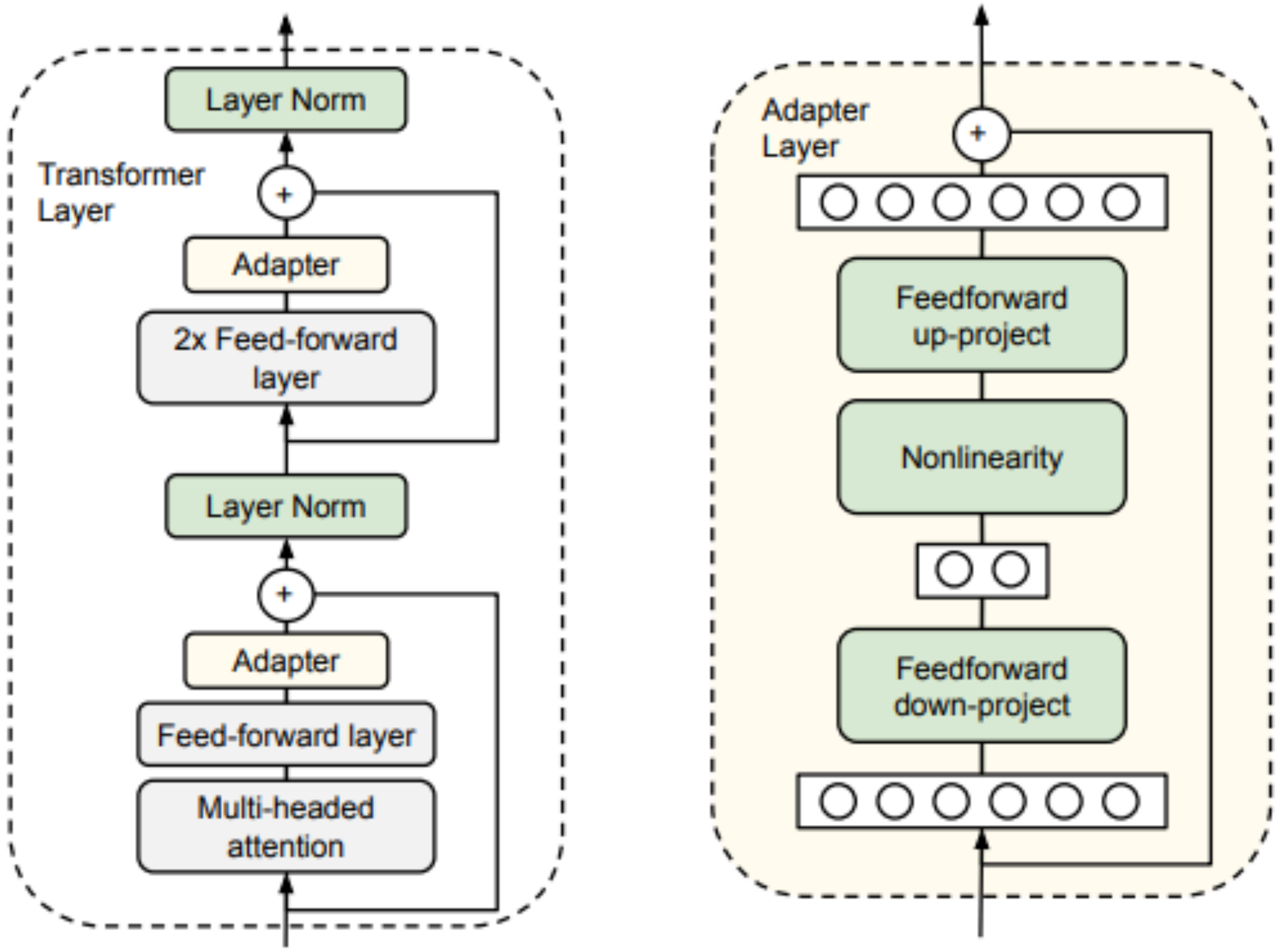
Slide Credit: Stanford CS224n, John Hewitt

[Liu et al., 2019; Joshi et al., 2020]

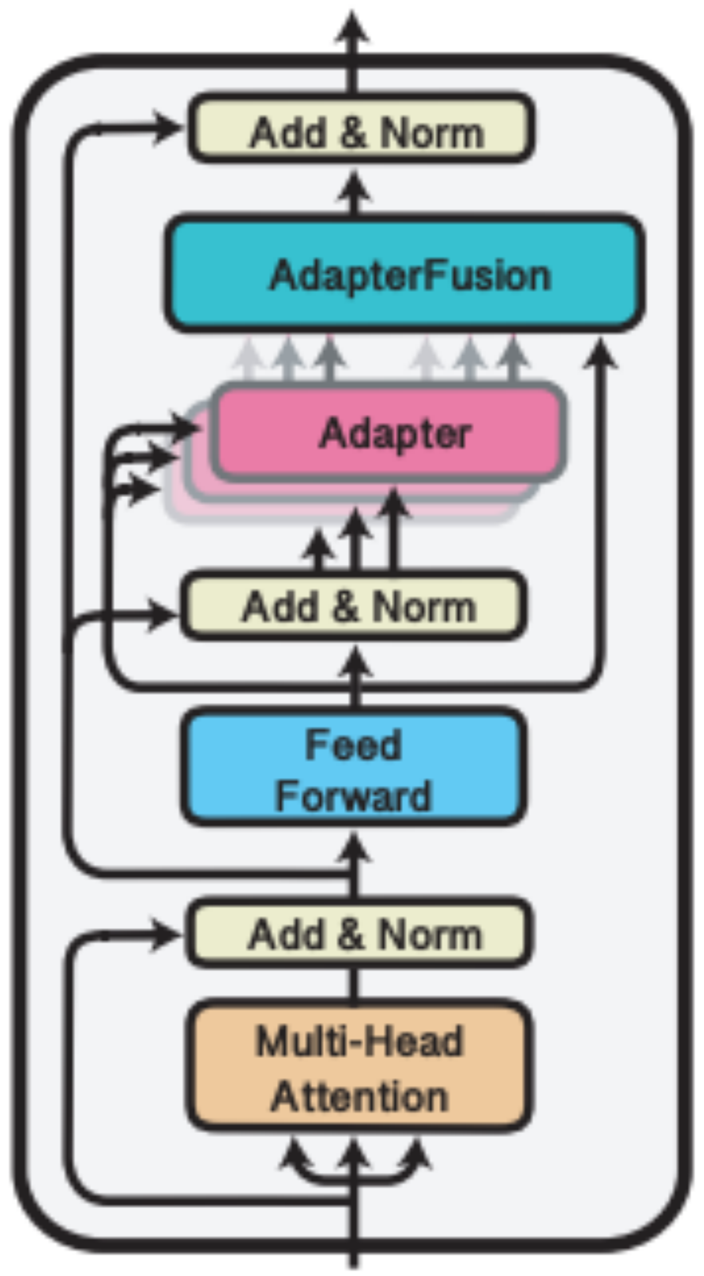
Parameter-Efficient Finetuning: Adapters

- Add lightweight network with new learnable parameters
- Only these parameters are fine-tuned, rest are frozen





[Houlsby et al., 2019]

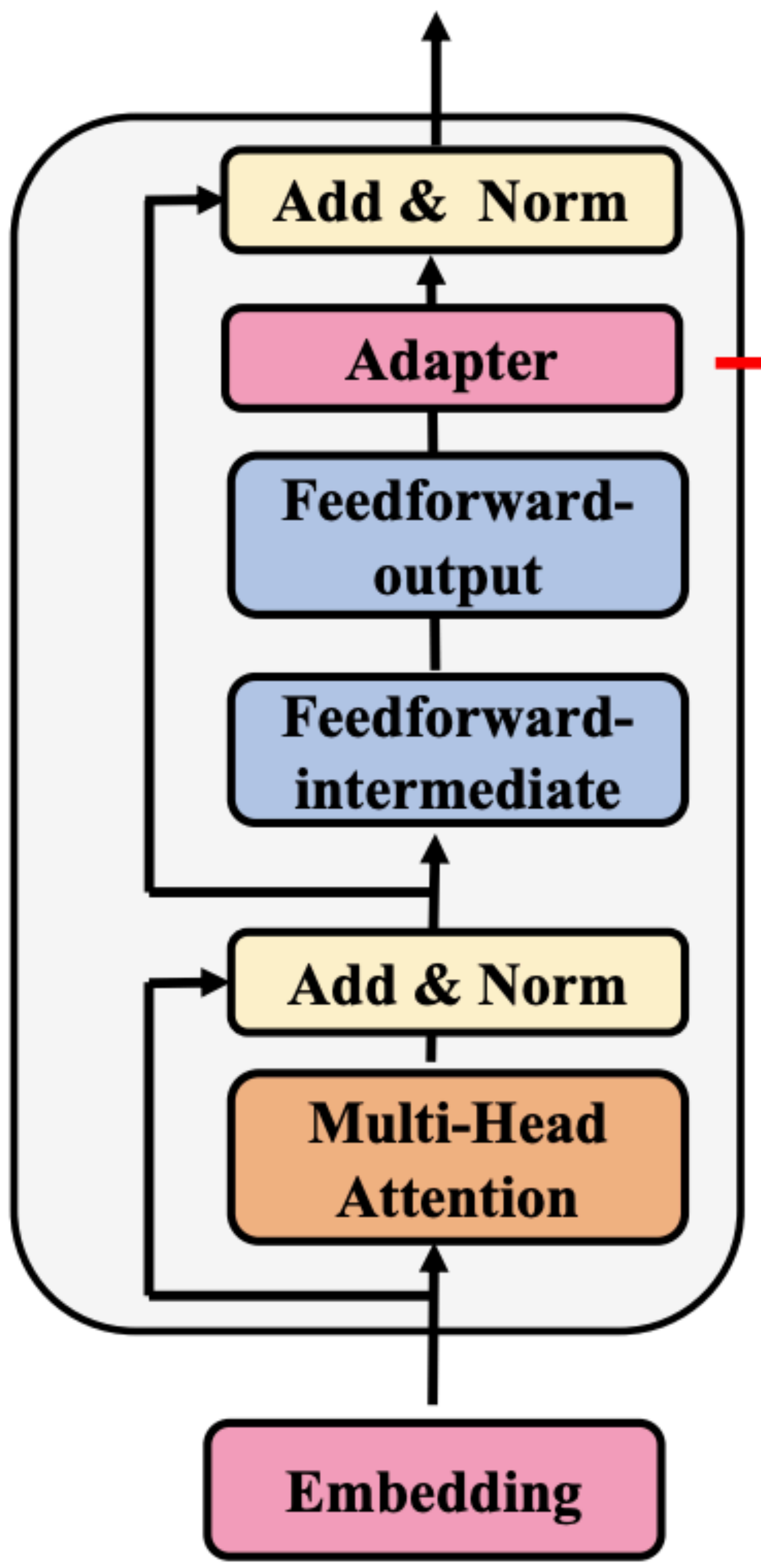


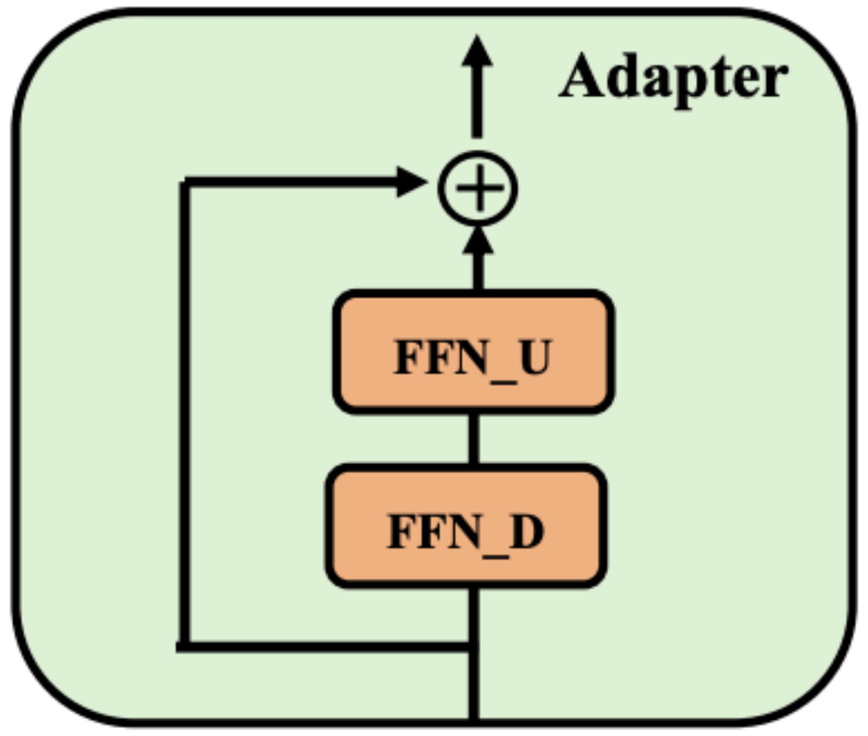
[Pfeiffer et al., 2021]

https://github.com/adapter-hub/adapter-transformers

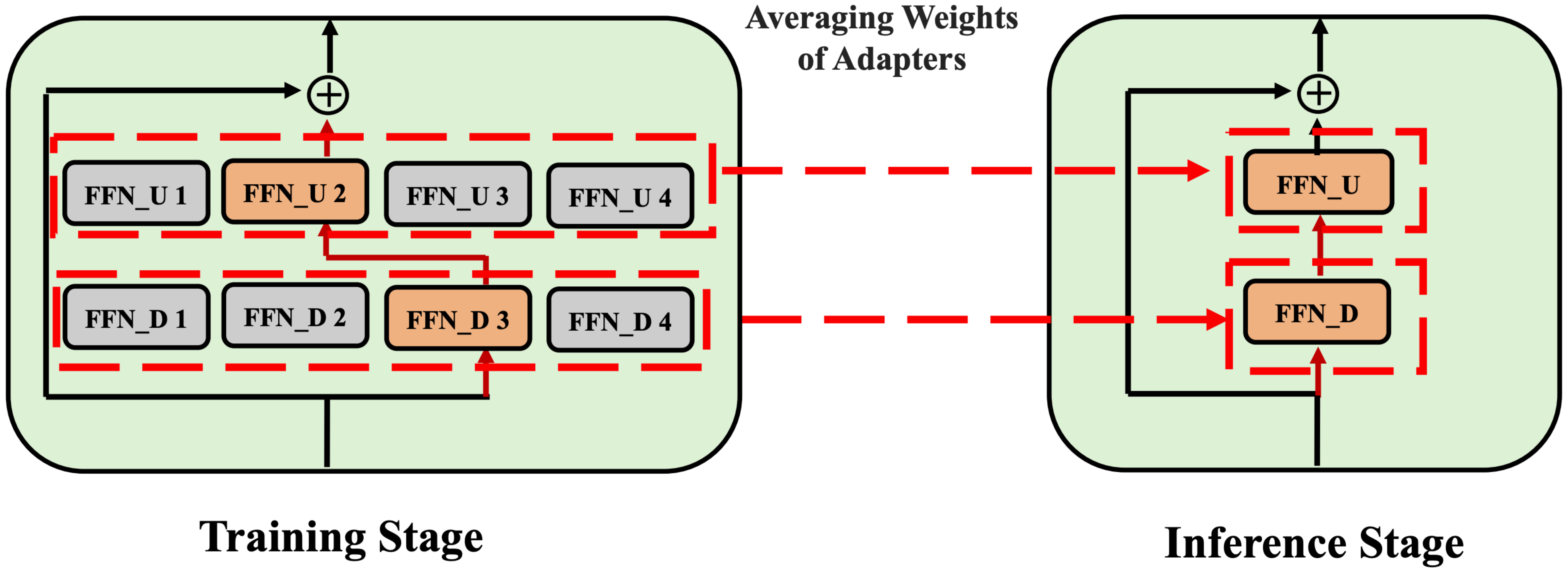
Parameter-Efficient Finetuning: Adapters

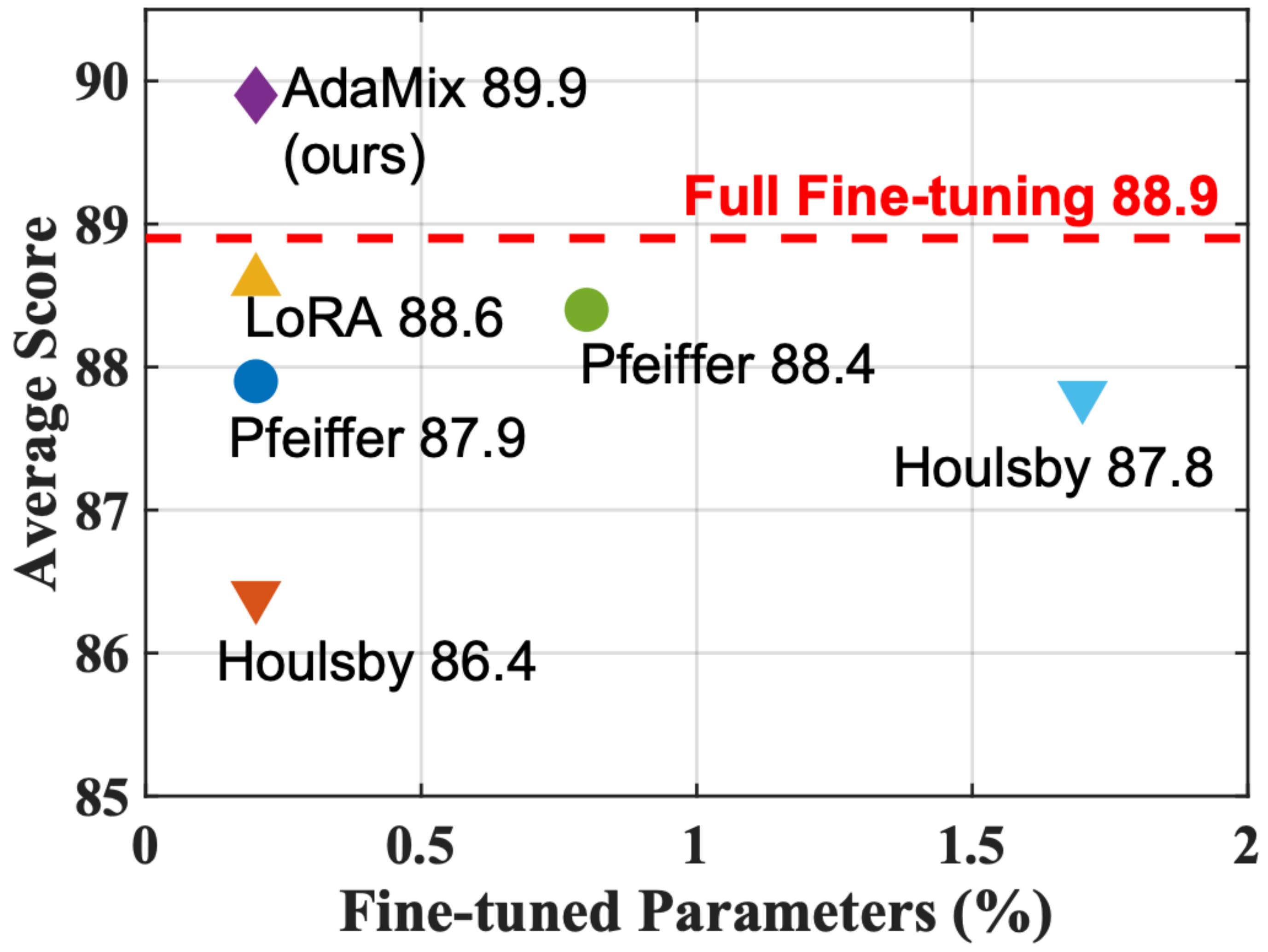
- **Mixture of adapters** - stochastically selected during training
- Average weights of adapters during inference





[Houlsby et al., 2019]



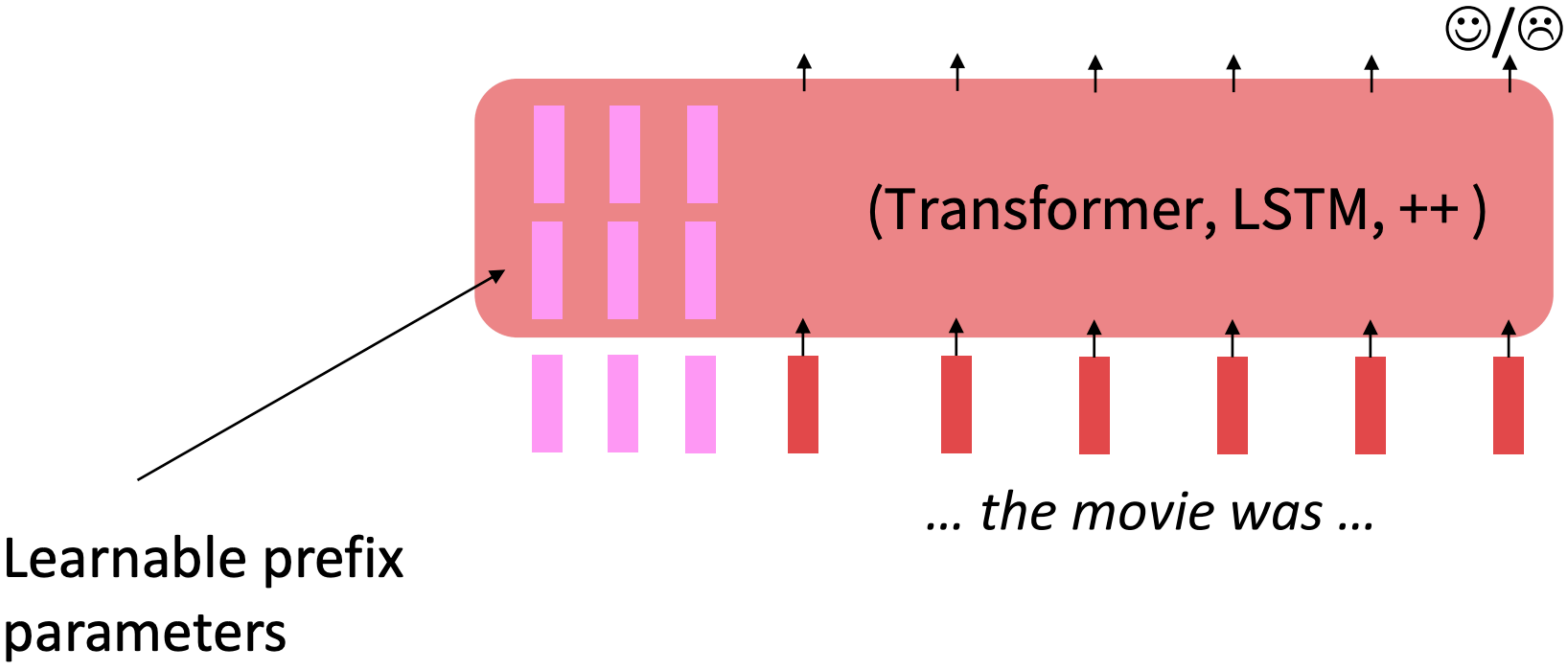


Performance on GLUE, fine-tuning of RoBERTa-large

[AdaMix, Wang et al., EMNLP 2022]

Parameter-Efficient Finetuning: Prefix-Tuning, Prompt tuning

- Prefix-Tuning adds a prefix of parameters, and freezes all pretrained parameters.
- The prefix is processed by the model just like real words would be.
- Advantage: each element of a batch at inference could run a different tuned model.

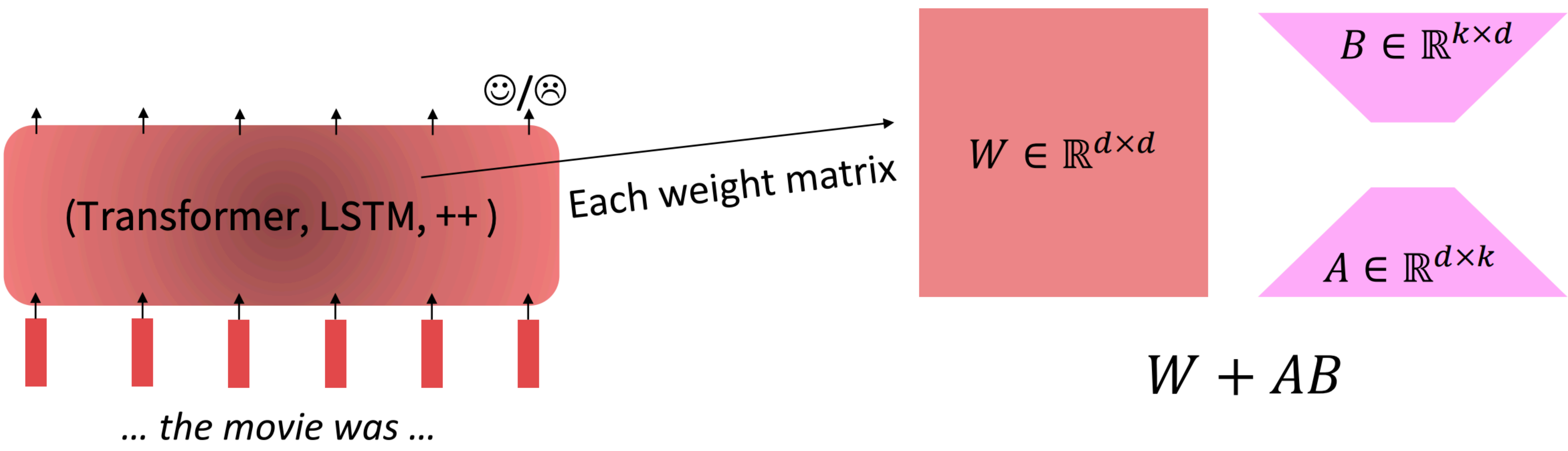


Slide Credit: Stanford CS224n, John Hewitt

[Li and Liang, 2021; Lester et al., 2021]

Parameter-Efficient Finetuning: Low-Rank Adaptation

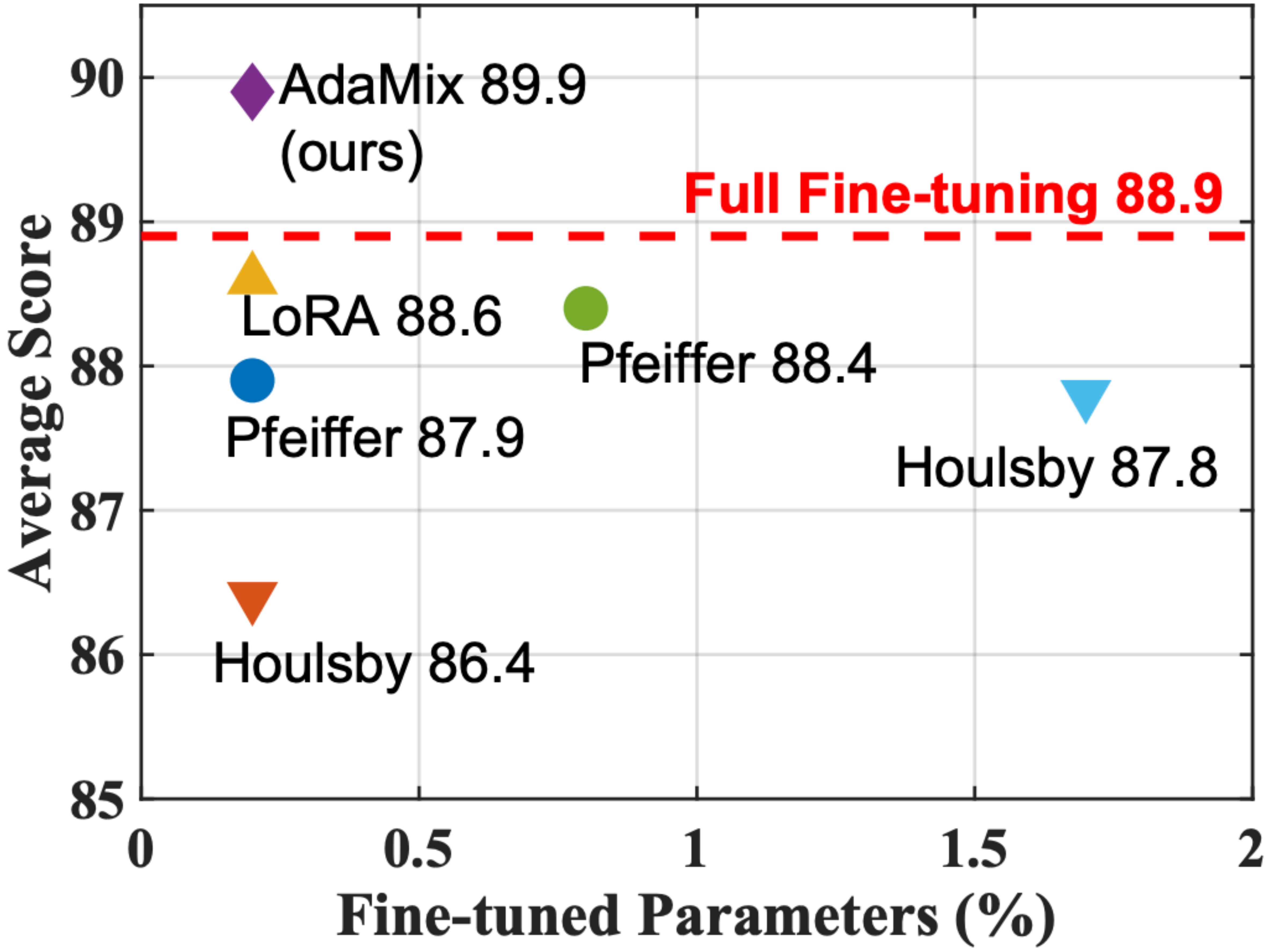
- Low-Rank Adaptation Learns a low-rank "diff" between the pretrained and finetuned weight matrices.
- Easier to learn than prefix-tuning



[Hu et al., 2021]

Parameter-Efficient Finetuning: Low-Rank Adaptation

Model	#Param.	M A	…	S-B arson	Avg.
Full Fine-tuning†	355.0M	9		**2.4**	88.9
Pfeiffer Adapter†	3.0M	9		2.1	88.4
Pfeiffer Adapter†	0.8M	9		1.9	87.9
Houlsby Adapter†	6.0M	8		1.0	87.8
Houlsby Adapter†	0.8M	9		1.5	86.4
LoRA†	0.8M	9		2.3	88.6
AdaMix Adapter	0.8M	**9**		**2.4**	**89.9**



Good performance by tuning just a fraction of the weights

[AdaMix, Wang et al., EMNLP 2022]

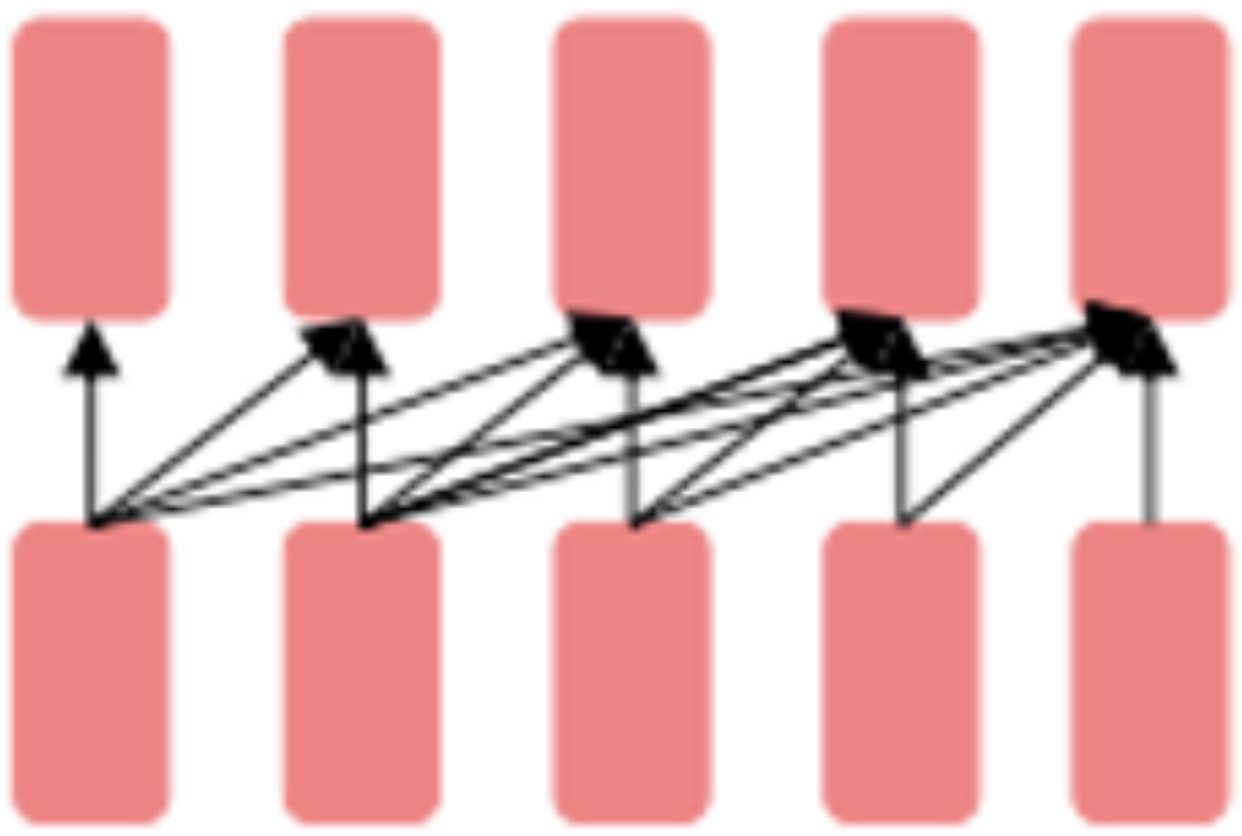
Transformers for pertaining

- Self-supervised Transformer based models shattered language understanding benchmarks in NLP in 2018.
- Trained on large text corpus with self-supervised objectives and then transferred.

Encoder only

- Masked language models
- Bidirectional context
- BERT + variants (e.g. RoBERTa)

Decoder only



- Language models
- Can't condition on future words, good for generation
- GPT-2, GPT-3, LaMDA, PaLM

Encoder-Decoder

- Combine benefits of both
- Original Transformer, UniLM, BART, T5, Meena

GPT models

GPT

- Improving language understanding by generative pre-training (Radford et al, 2018)
- Large language model with transformers with fine-tuning!
- Trained on BooksCorpus (800M words), 117M parameters (12 layers)

GPT-2

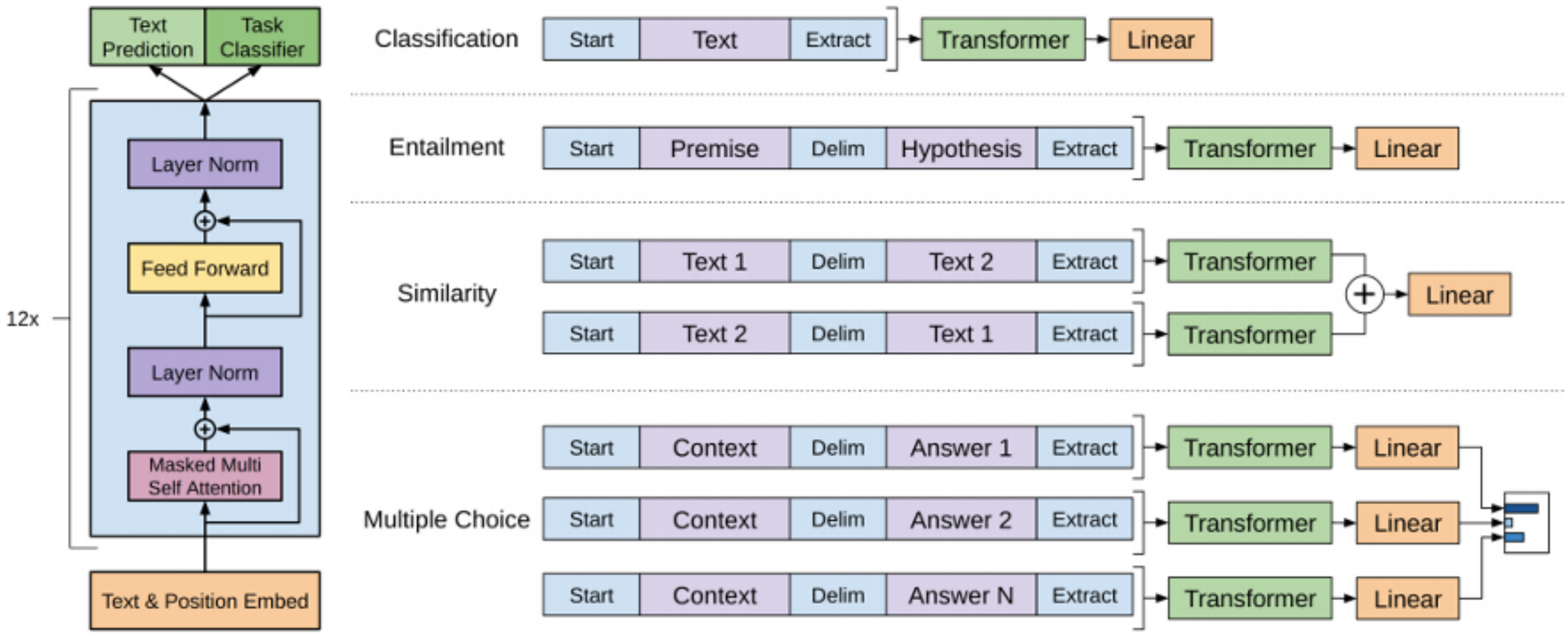
- Language Models are Unsupervised Multitask Learner (Radford et al, 2019)
- Trained on WebText (40B words), 1.5B parameters (48 layers)
- No fine-tuning, few-shot learning

GPT-3

- Language Models are Few-Shot Learners (Brown et al, 2020)
- Trained on Web+Books+Wikipedia (300B words), 175B parameters (96 layers)

GPT (Generative pretrained transformer)

- Unsupervised retraining: Standard language model loss
- Supervised fine-tuning: Use simple classifier (linear layer + softmax) trained to predict correct class (use combined loss)



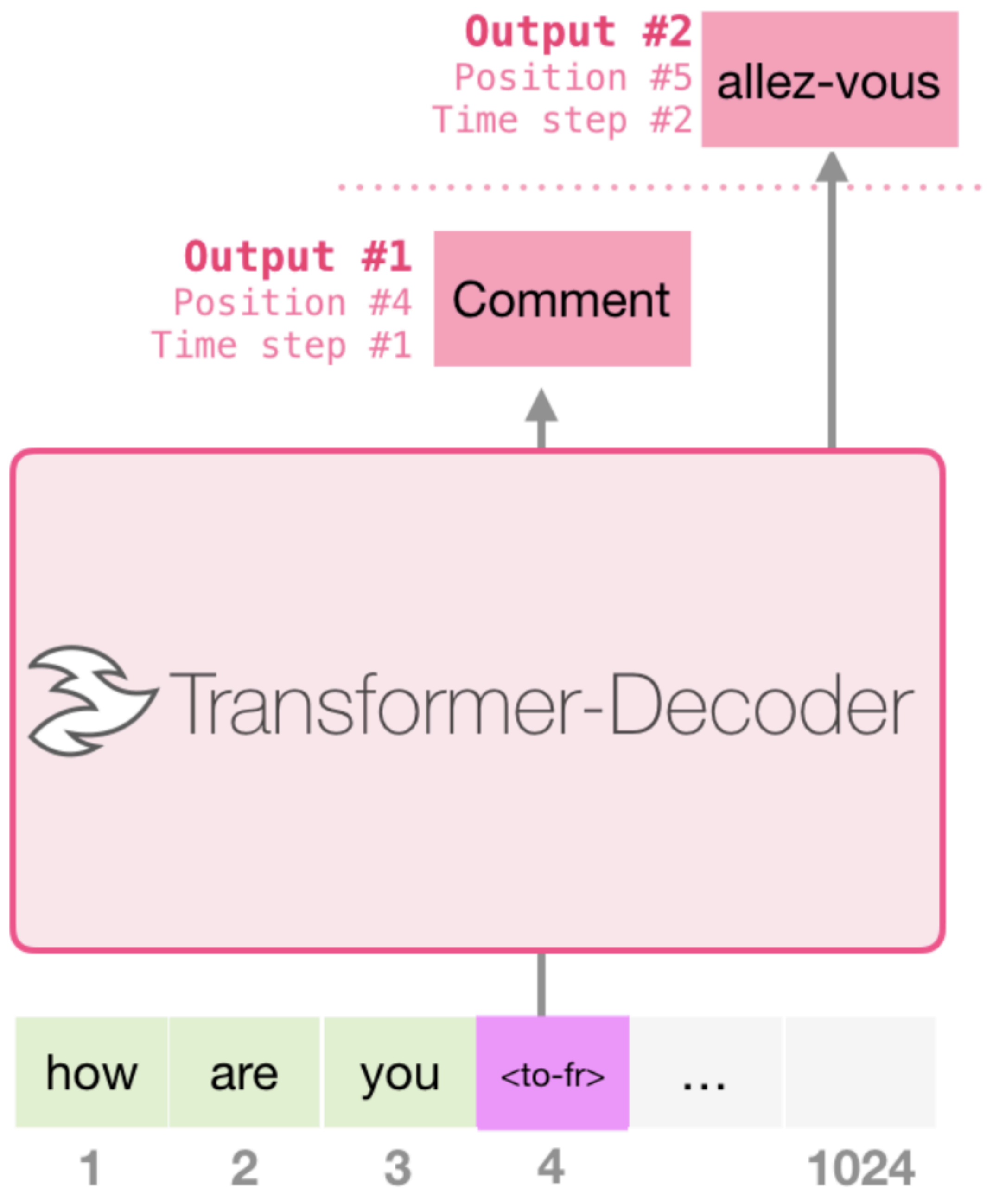
Improving language understanding by generative pre-training (Radford et al, 2018)

GPT-2

- Express all tasks a a language modelling task
- Training improvements
  - Improved initialization / additional layer normalization
  - Increased vocabulary / context /batch size

- Machine Translation

I	am	a	student	<to-fr>	je	suis	étudiant
let	them	eat	cake	<to-fr>	Qu'ils	mangent	de
good	morning	<to-fr>	Bonjour				

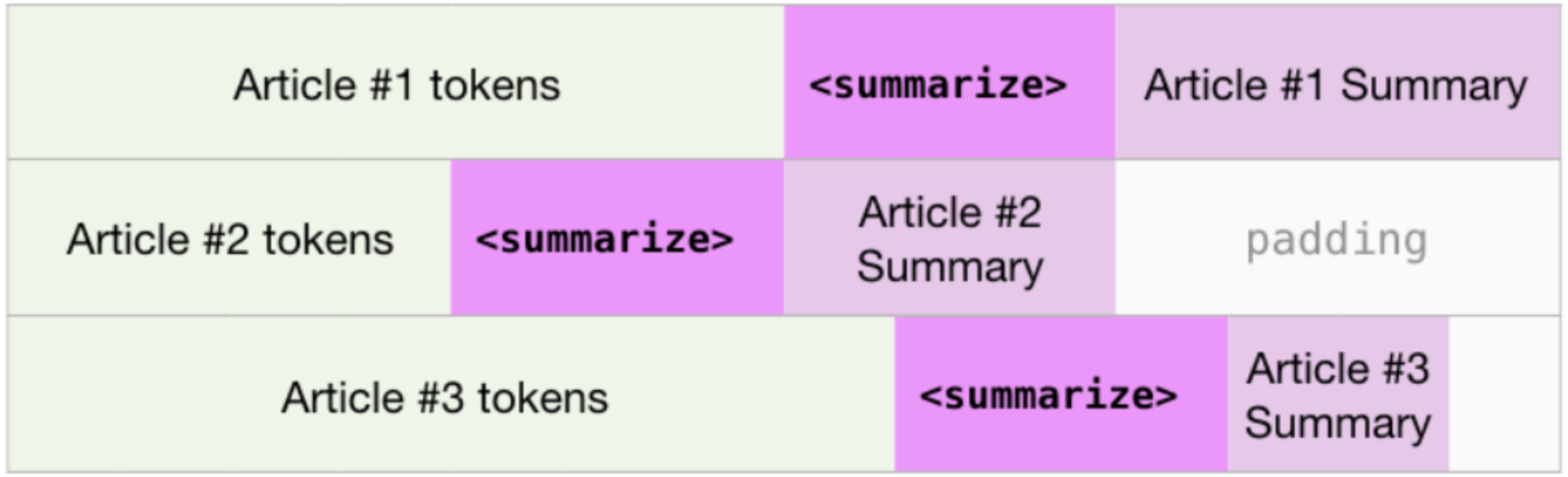


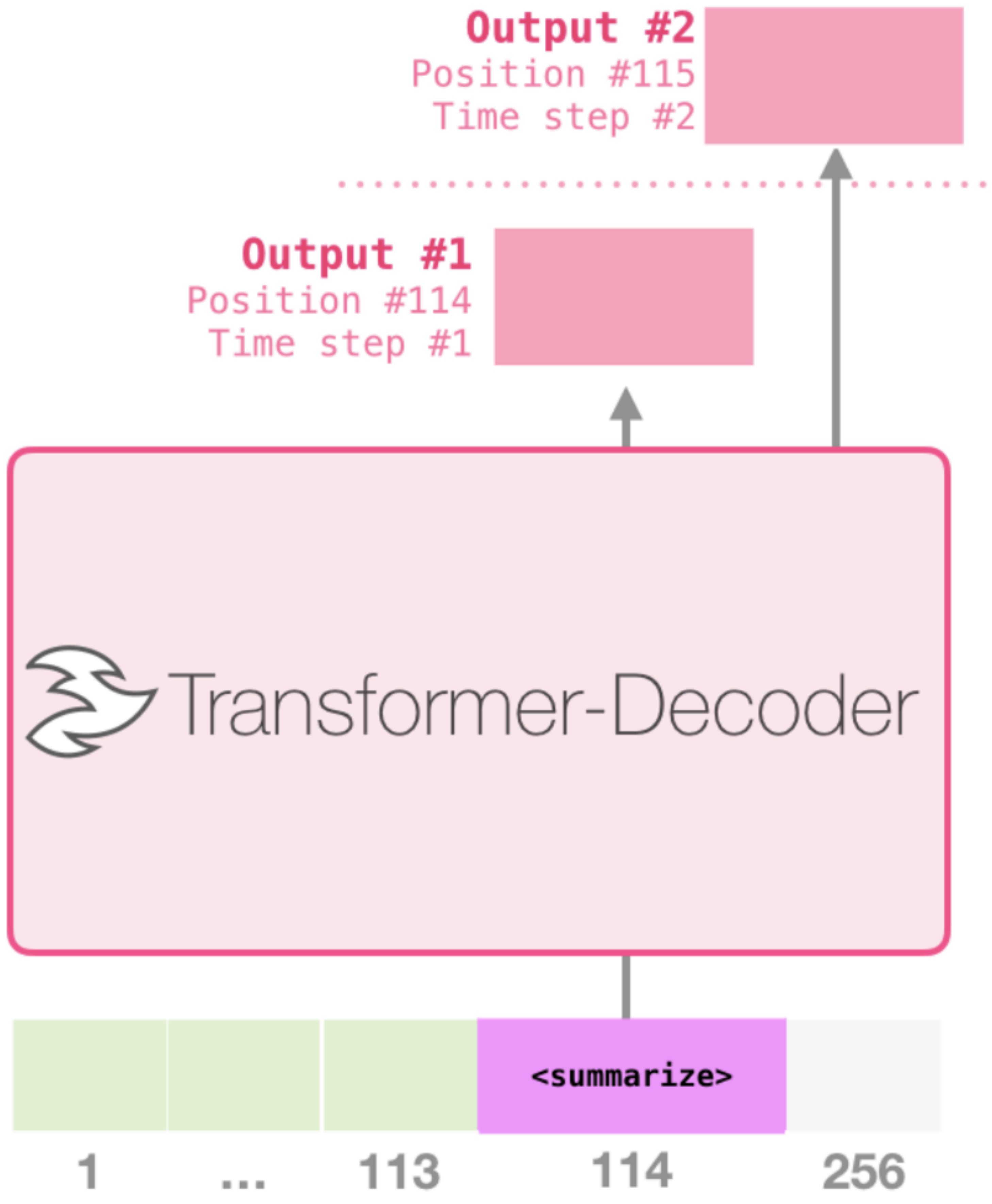
(figure credit: Jay Alammar http://jalammar.github.io/illustrated-gpt2/)

GPT-2

How can we use decoders for different tasks?

- Summarization





(figure credit: Jay Alammar
http://jalammar.github.io/illustrated-gpt2/)

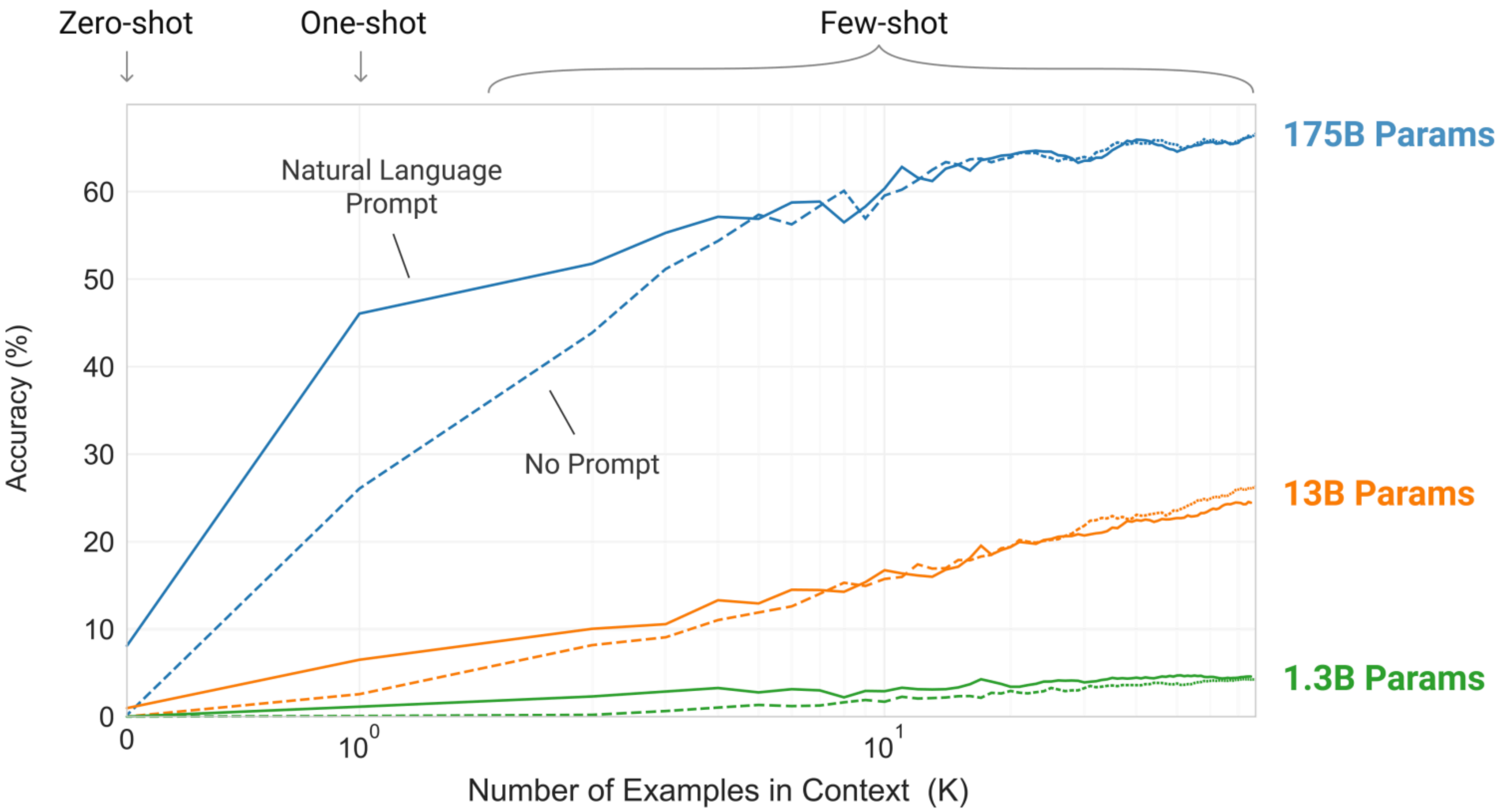
Language Models are Unsupervised Multitask Learner (Radford et al. 2019)

GPT-3: Few-shot learning

0 training examples

1 training examples

A few examples are provided at test time



Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
Translate English to French:      ← task description
cheese =>                         ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
Translate English to French:      ← task description
sea otter => loutre de mer        ← example
cheese =>                         ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
Translate English to French:      ← task description
sea otter => loutre de mer        ← examples
peppermint => menthe poivrée
plush girafe => girafe peluche
cheese =>                         ← prompt
```

Language Models are Few-Shot Learners (Brown et al. OpenAI, 2020)

Transformers for pretraining

- Self-supervised Transformer based models shattered language understanding benchmarks in NLP in 2018.
- Trained on large text corpus with self-supervised objectives and then transferred.

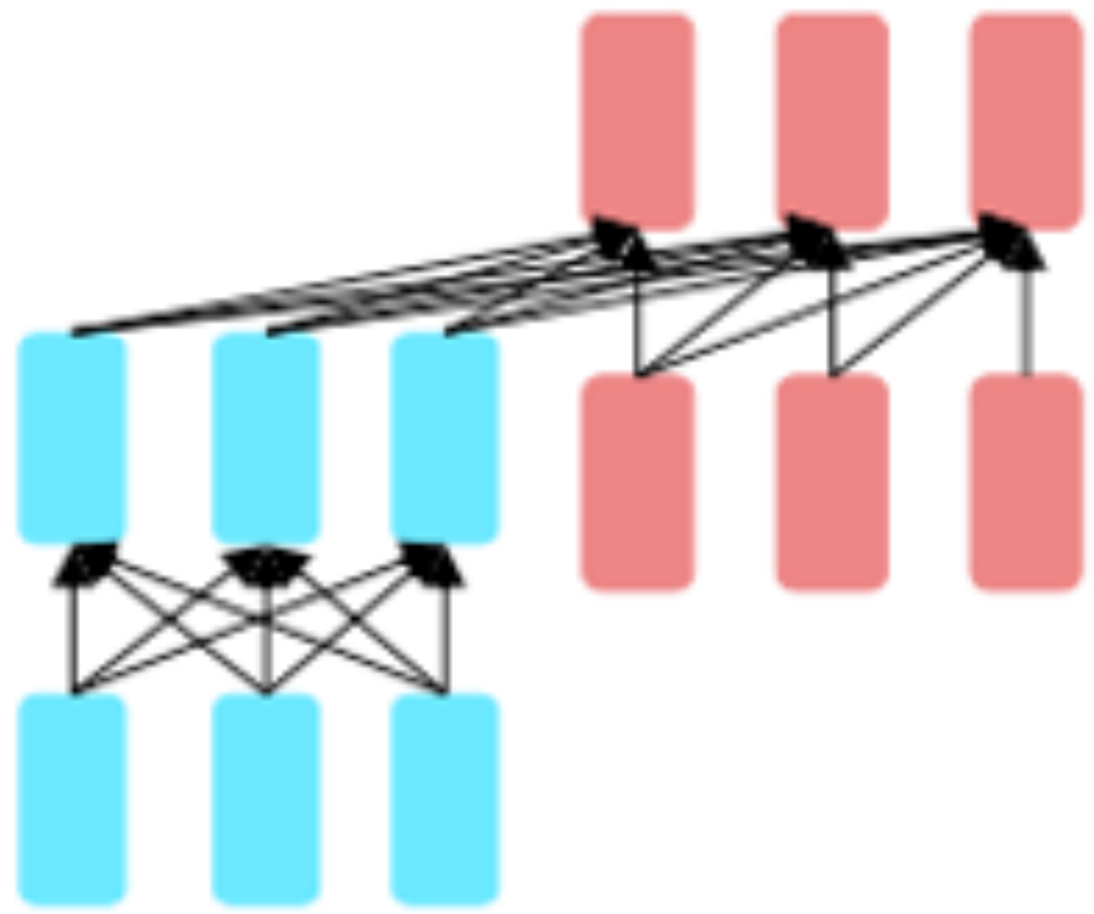
Encoder only

- Masked language models
- Bidirectional context
- BERT + variants (e.g. RoBERTa)

Decoder only

- Language models
- Can’t condition on future words, good for generation
- GPT-2, GPT-3, LaMDA

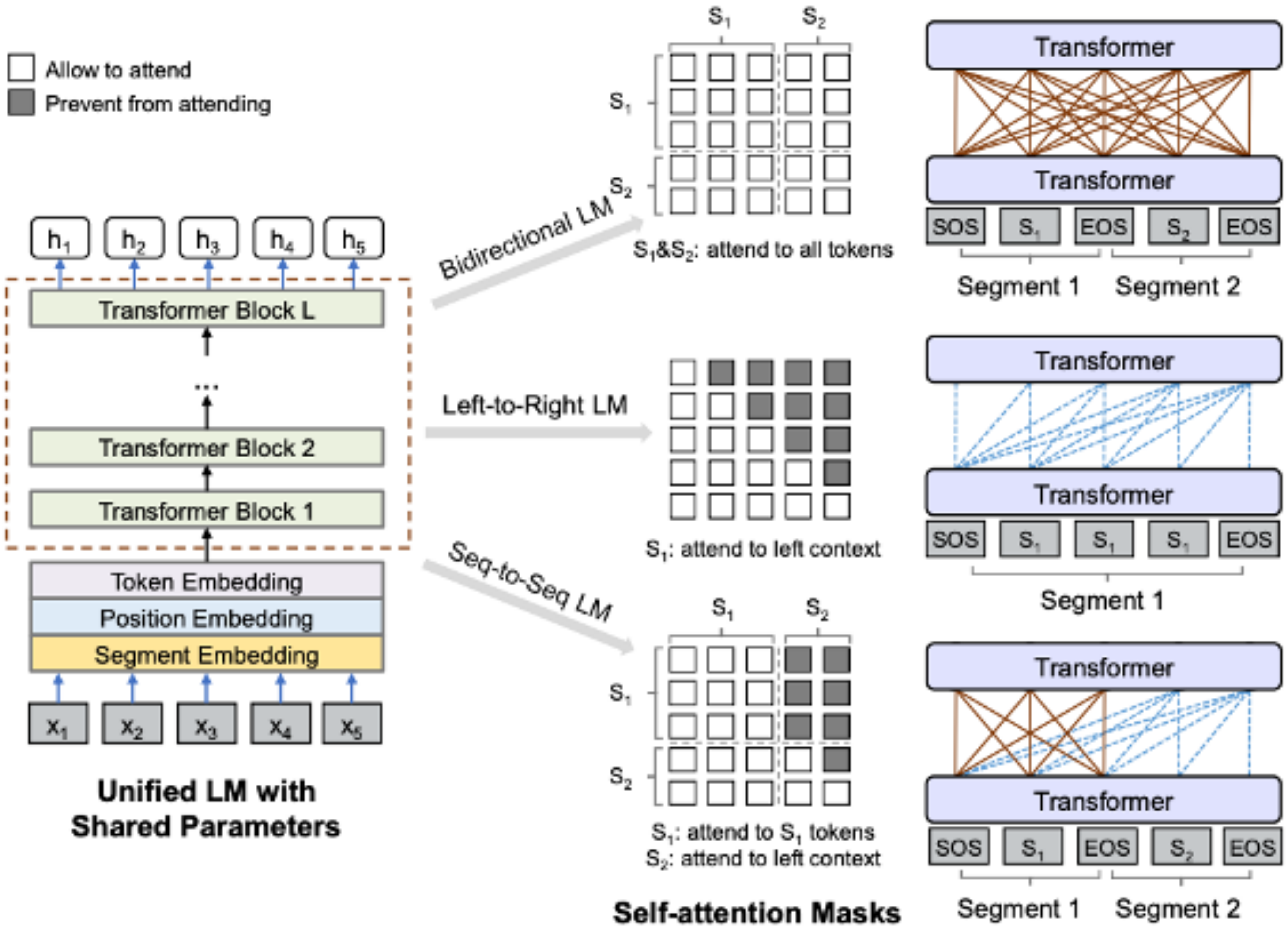
Encoder-Decoder



- Combine benefits of both
- Original Transformer, UniLM, BART, T5, Meena

Encoder-Decoder pretraining

- Combine advantages of both encoder and decoder
- Seq2Seq LM with masking
- Next sentence prediction



Unified Language Model Pre-training for Natural Language Understanding and Generation [Dong et al, NeurIPS 2019]

UniLM v1

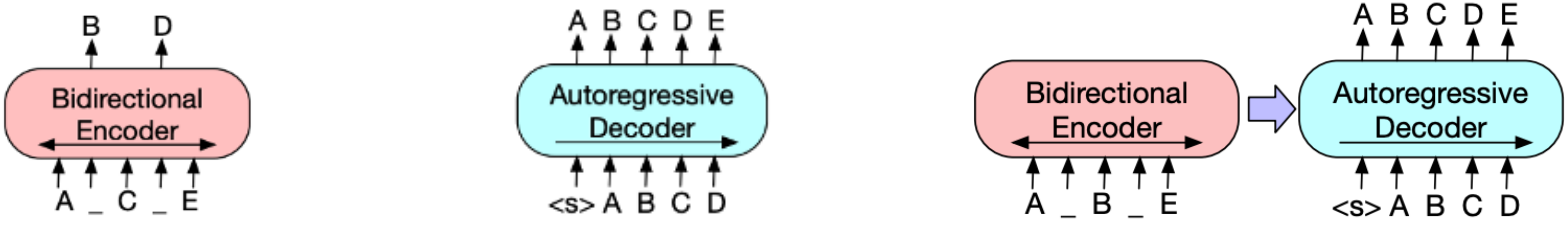
- Combine benefits of BERT (encoder) and GPT (decoder)

Model	CoLA MCC	SST-2 Acc	MRPC F1	STS-B S Corr	QQP F1	MNLI-m/mm Acc	QNLI Acc	RTE Acc	WNLI Acc	AX Acc	**Score**
GPT	45.4	91.3	82.3	80.0	70.3	82.1/81.4	87.4	56.0	53.4	29.8	72.8
$\text{BERT}_{\text{LARGE}}$	60.5	**94.9**	89.3	86.5	**72.1**	86.7/**85.9**	**92.7**	70.1	65.1	39.6	80.5
UNILM	**61.1**	94.5	**90.0**	**87.7**	71.7	**87.0/85.9**	**92.7**	**70.9**	65.1	38.4	**80.8**

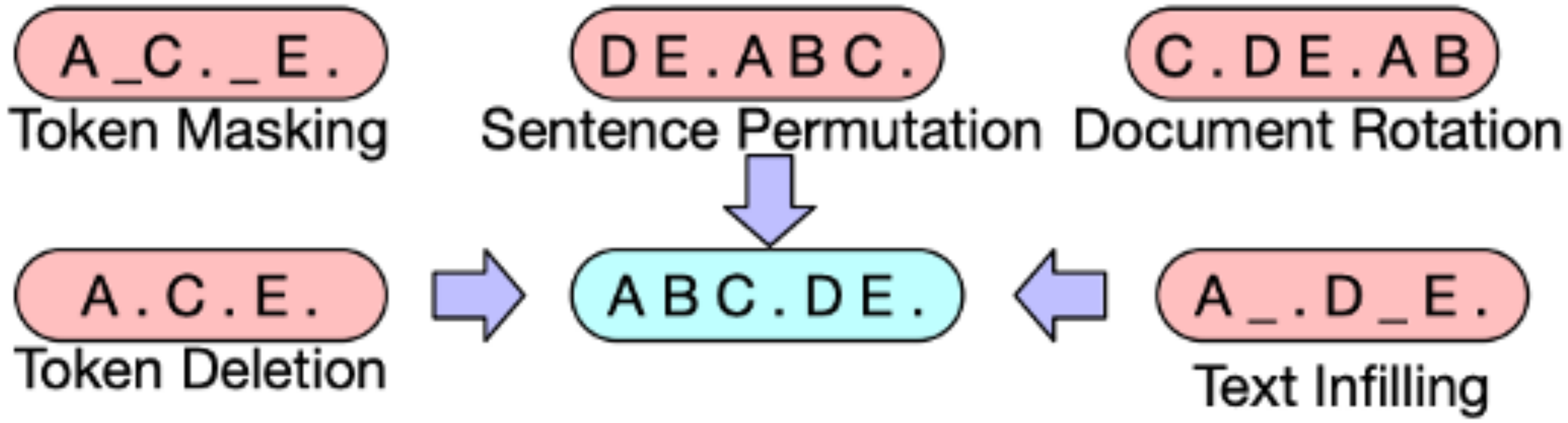
Unified Language Model Pre-training for Natural Language Understanding and Generation
Dong et al, Microsoft, NeurIPS 2019

BART: Denoising seq2seq training

BERT GPT BART



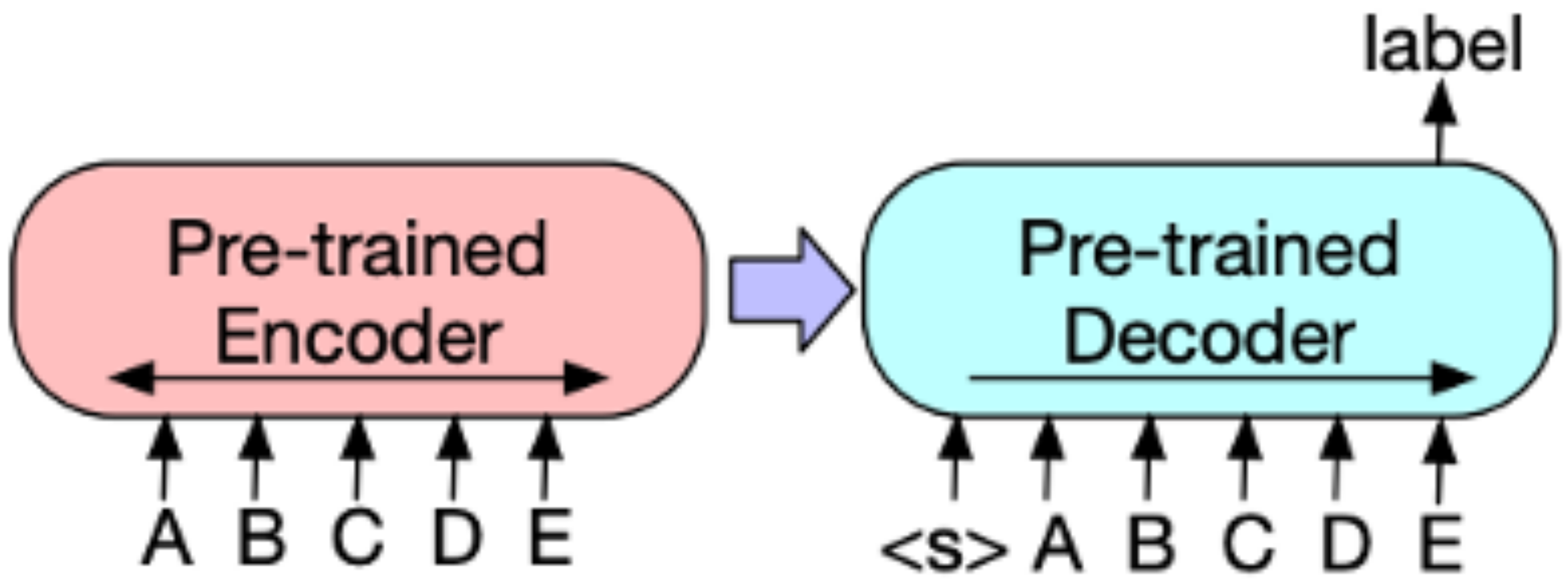
- Combine benefits of BERT (encoder) and GPT (decoder)
- More flexibility in noise generation



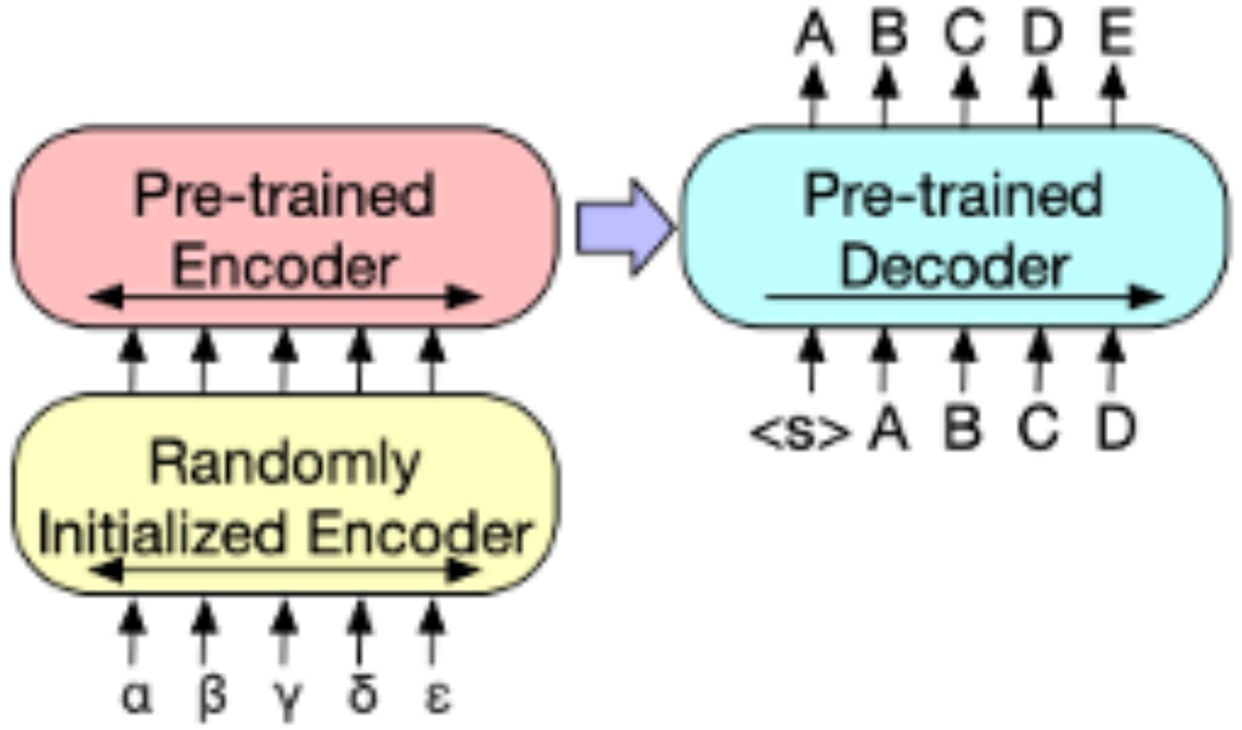
BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension
Lewis et al, Facebook AI, ACL 2020

BART: Denoising seq2seq training

Classification



Machine Translation

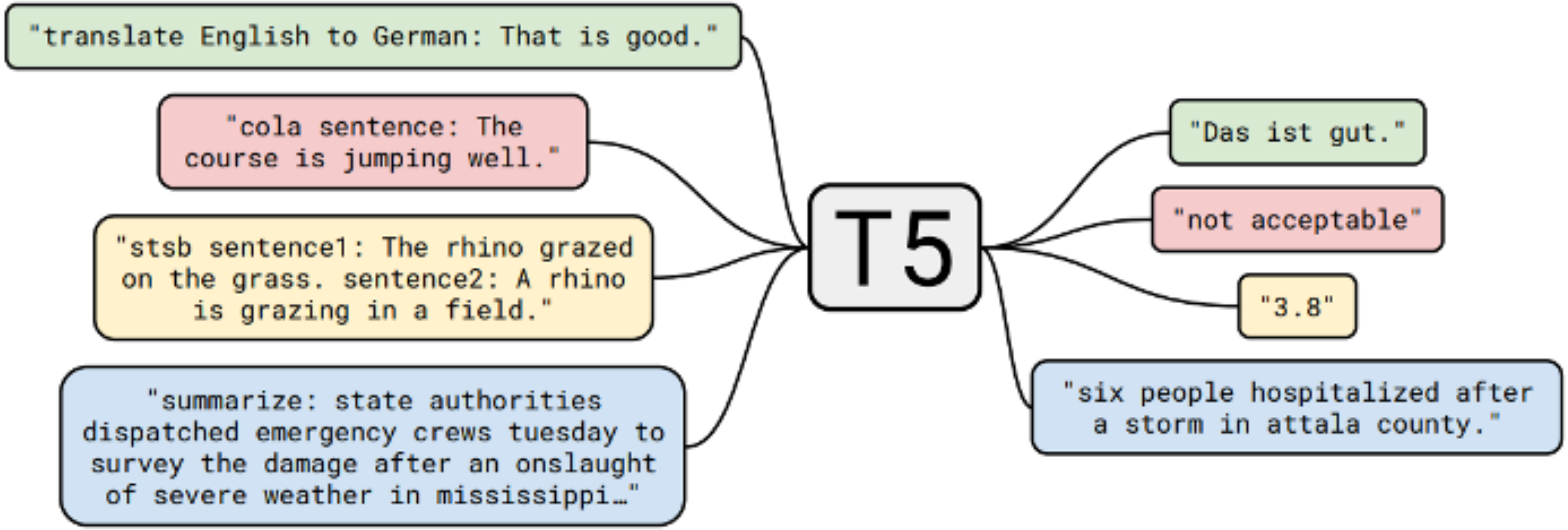


	SQuAD 1.1 EM/F1	SQuAD 2.0 EM/F1	MNLI m/mm	SST Acc	QQP Acc	QNLI Acc	STS-B Acc	RTE Acc	MRPC Acc	CoLA Mcc
BERT	84.1/90.9	79.0/81.8	86.6/-	93.2	91.3	92.3	90.0	70.4	88.0	60.6
UniLM	-/-	80.5/83.4	87.0/85.9	94.5	-	92.7	-	70.9	-	61.1
XLNet	**89.0**/94.5	86.1/88.8	89.8/-	95.6	91.8	93.9	91.8	83.8	89.2	63.6
RoBERTa	88.9/**94.6**	**86.5/89.4**	**90.2/90.2**	96.4	92.2	94.7	**92.4**	86.6	**90.9**	**68.0**
BART	88.8/**94.6**	86.1/89.2	89.9/90.1	**96.6**	**92.5**	**94.9**	91.2	**87.0**	90.4	62.8

BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension
Lewis et al, Facebook AI, ACL 2020

T5: Text to Text Transfer Transformer

- Treat all NLP problems as encoding text and generating text
- Trained on cleaned up version of Common Crawl



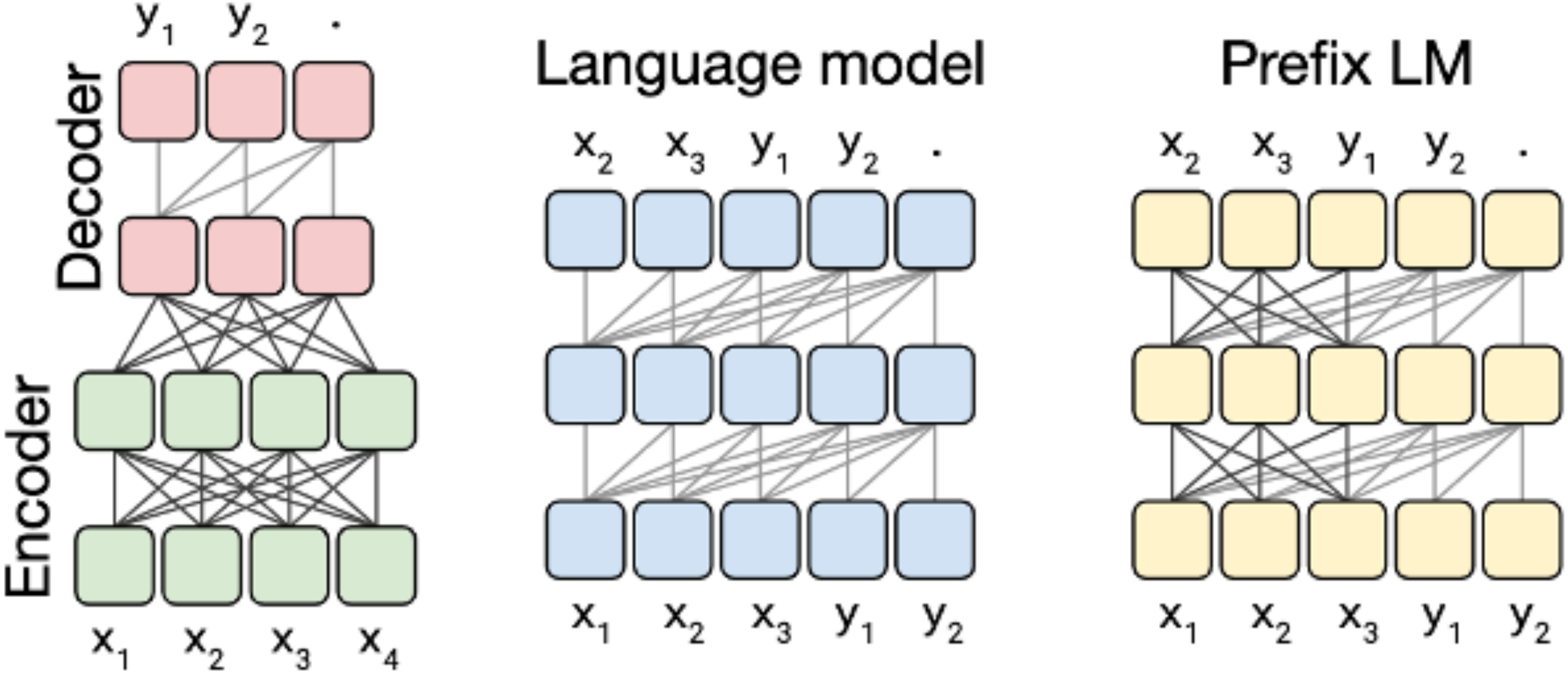
Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer [Raffel et al, Google, JMLR 2020]

T5: Text to Text Transfer Transformer

Normally: Separate parameters for encoder/decoder

Causal masking only

Masking similar to encoder/decoder

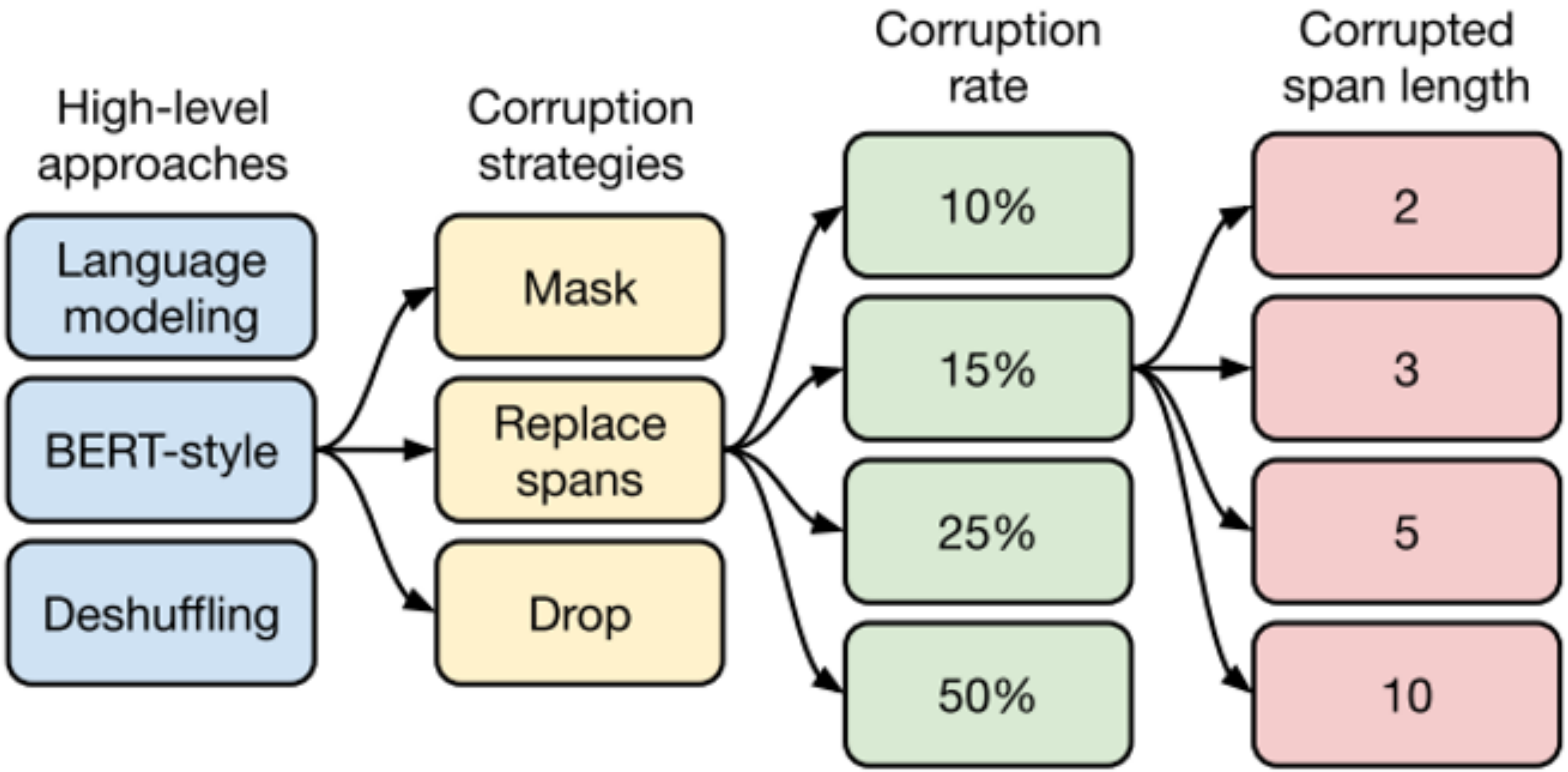


Can force sharing of parameters for encoder/decoder

Similar performance, less parameters

Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer [Raffel et al, Google, JMLR 2020]

T5: Text to Text Transfer Transformer

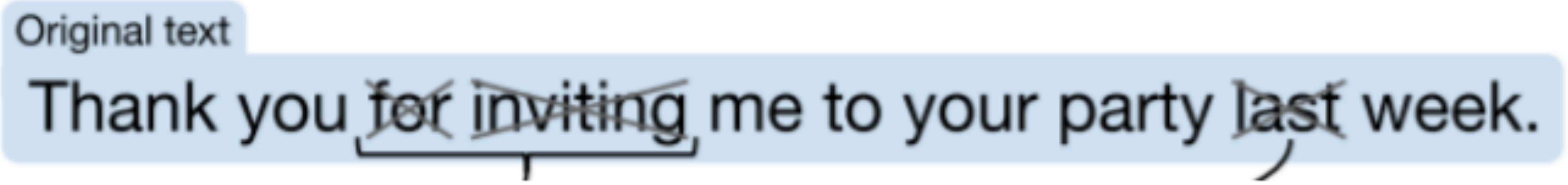


Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer [Raffel et al, Google, JMLR 2020]

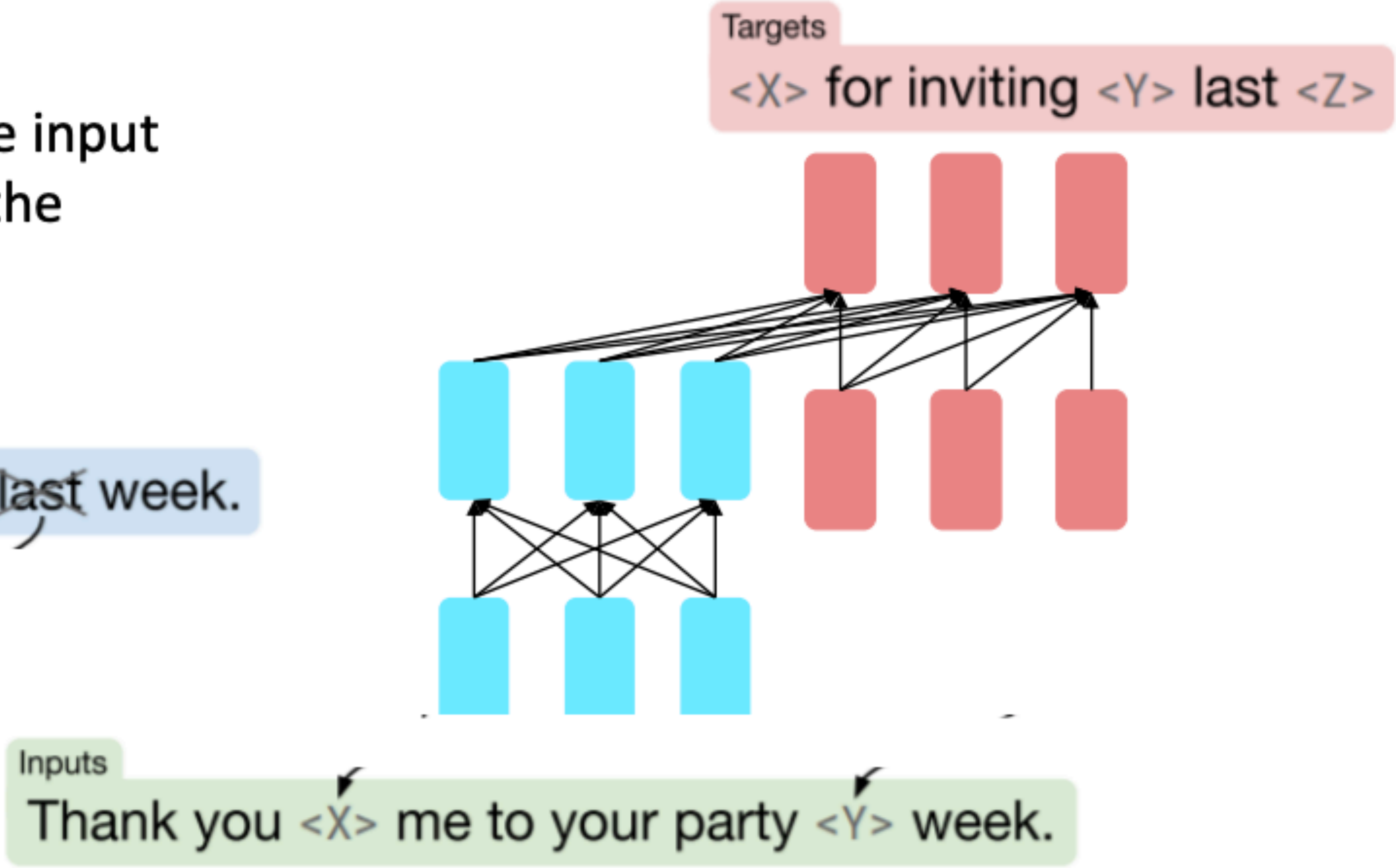
T5 (use both encoder and decoder)

Span corruption works best

Replace different-length spans from the input with unique placeholders; decode out the spans that were removed!



This is implemented in text preprocessing: it’s still an objective that looks like **language modeling** at the decoder side.



Slide Credit: Stanford CS224n, John Hewitt

T5: Text to Text Transfer Transformer

Different corruption type

	Objective	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
Predict all	BERT-style (Devlin et al., 2018)	82.96	19.17	**80.65**	69.85	26.78	**40.03**	27.41
	MASS-style (Song et al., 2019)	82.32	19.16	80.10	69.28	26.79	**39.89**	27.55
Predict corrupted	★ Replace corrupted spans	83.28	**19.24**	**80.88**	**71.36**	**26.98**	39.82	**27.65**
	Drop corrupted tokens	**84.44**	**19.31**	**80.52**	68.67	**27.07**	39.76	**27.82**

Different corruption rate

Corruption rate	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
10%	**82.82**	19.00	**80.38**	69.55	**26.87**	39.28	**27.44**
★ 15%	**83.28**	19.24	**80.88**	**71.36**	**26.98**	**39.82**	**27.65**
25%	**83.00**	**19.54**	**80.96**	70.48	**27.04**	**39.83**	**27.47**
50%	81.27	19.32	79.80	70.33	**27.01**	**39.90**	**27.49**

Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer [Raffel et al, Google, JMLR 2020]

T5 (use both encoder and decoder)

Raffel et al., 2018 found encoder-decoders to work better than decoders for their tasks, and span corruption (denoising) to work better than language modeling.

Architecture	Objective	Params	Cost	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
★ Encoder-decoder	Denoising	$2P$	M	**83.28**	**19.24**	**80.88**	**71.36**	**26.98**	**39.82**	**27.65**
Enc-dec, shared	Denoising	P	M	82.81	18.78	**80.63**	**70.73**	26.72	39.03	**27.46**
Enc-dec, 6 layers	Denoising	P	$M/2$	80.88	18.97	77.59	68.42	26.38	38.40	26.95
Language model	Denoising	P	M	74.70	17.93	61.14	55.02	25.09	35.28	25.86
Prefix LM	Denoising	P	M	81.82	18.61	78.94	68.11	26.43	37.98	27.39
Encoder-decoder	LM	$2P$	M	79.56	18.59	76.02	64.29	26.27	39.17	26.86
Enc-dec, shared	LM	P	M	79.60	18.13	76.35	63.50	26.62	39.17	27.05
Enc-dec, 6 layers	LM	P	$M/2$	78.67	18.26	75.32	64.06	26.13	38.42	26.89
Language model	LM	P	M	73.78	17.54	53.81	56.51	25.23	34.31	25.38
Prefix LM	LM	P	M	79.68	17.84	76.87	64.86	26.28	37.51	26.76

Slide Credit: Stanford CS224n, John Hewitt